# 1 Fundamentals: Logic, sets, numbers and vectors

## 1.1 Real Numbers, Eulidean Spaces

**Axioms:**

• **Axioms of addition:**

| | | |
|---|---|---|
| **A1 (Assoc.)** | $x+(y+z)=(x+y)+z$ | $\forall x,y,z\in\mathbb{R}$ |
| **A2 (Neutral el.)** | $x+0=x$ | $\forall x\in\mathbb{R}$ |
| **A3 (Inverse el.)** | $x+(-x)=0$ | $\forall x\in\mathbb{R}\ \exists(-x)\in\mathbb{R}$ |
| **A4 (Commut.)** | $x+y=y+x$ | $\forall x,y\in\mathbb{R}$ |

**Axioms of Multiplication:**

| | | |
|---|---|---|
| **M1 (Assoc.)** | $x\cdot(y\cdot z)=(x\cdot y)\cdot z$ | $\forall x,y,z\in\mathbb{R}$ |
| **M2 (Neutral el.)** | $x\cdot 1=x$ | $\forall x\in\mathbb{R}$ |
| **M3 (Inverse el.)** | $x\cdot x^{-1}=1$ | $\forall x\in\mathbb{R}\setminus\{0\}\ \exists x^{-1}\in\mathbb{R}$ |
| **M4 (Commut.)** | $x\cdot z=z\cdot x$ | $\forall x,z\in\mathbb{R}$ |

**Order Axioms:**

| | | |
|---|---|---|
| **O1 (Reflexivity)** | $x\le x$ | $\forall x\in\mathbb{R}$ |
| **O2 (Transitivity)** | $(x\le y\wedge y\le z)\Rightarrow x\le z$ | |
| **O3 (Antisymmetry)** | $(x\le y\wedge y\le x)\Rightarrow x=y$ | |
| **O4 (Totality)** | $\forall x,y\in\mathbb{R}:\ x\le y$ or $y\le x$ | |

**Distributivity Axiom:**

$$x\cdot(y+z)=xy+xz=(y+z)\cdot x\qquad\forall x,y,z\in\mathbb{R}$$

**Compatibility Axioms:**

| | | |
|---|---|---|
| **K1** | $x\le y\Rightarrow x+z\le y+z$ | $\forall x,y,z\in\mathbb{R}$ |
| **K2** | $x\ge 0,\ y\ge 0\Rightarrow xy\ge 0$ | $\forall x,y\in\mathbb{R}$ |

**R 1.1.5:**

• The set $\mathbb{Q}$ of rational numbers, equipped with addition, multiplication, and the order relation $\le$, satisfies the above axioms.

**V (Completeness axiom):**

• The set $\mathbb{R}$ is **order complete**: For any two nonempty sets $A,B\subset\mathbb{R}$ such that $a\le b$ for all $a\in A,\ b\in B$, there exists a number $c\in\mathbb{R}$ satisfying $a\le c\le b\quad\forall a\in A,\ b\in B$.

**C 1.1.6:**

• **The consequences of above axioms are:**

1. Uniqueness of the additive and multiplicative inverse.
2. $0\cdot x=0\quad\forall x\in\mathbb{R}$.
3. $(-1)\cdot x=-x\quad\forall x\in\mathbb{R}$, in particular $(-1)^2=1$.
4. $y\ge 0\iff(-y)\le 0$.
5. $y^2\ge 0\quad\forall y\in\mathbb{R}$, in particular $1=1\cdot 1\ge 0$.
6. If $x\le y$ and $u\le v$, then $x+u\le y+v$.
7. If $0\le x\le y$ and $0\le u$, then $xu\le yu$.

**T (Completeness of the real numbers):**

• $\mathbb{R}$ is a commutative ordered field that is order-complete.

**C 1.1.7 (Archimedean Principle):**

• 1. For $x\in\mathbb{R}$ and $y>0$ there exists $n\in\mathbb{N}$ such that $ny>x$.
• 2. For every $\varepsilon>0$ there exists $n\in\mathbb{N}$ such that $\frac{1}{n}<\varepsilon$.

**T 1.1.8:**

• For every $t\ge 0$, $t\in\mathbb{R}$, the equation $x^2=t$ has a solution in $\mathbb{R}$.

**D (Bounded set - upper/lower bounded):**

• A set $A\in\mathbb{R}$ is called upper bounded if $\exists b\in R\ \ \forall a\in A: a\le b$. Each such $b$ is called an upper bound for $A$. (Similarly for lower bound.)

**T (Supremum/Infimum):**

• 1. Maximum and minimum (if exists) are unique
2. Every non-empty subset bounded below (above) has a unique infimum (supremum).
3. $\sup(X)\ \Leftrightarrow\ (\forall x\in X: x\le S)\wedge(\forall\varepsilon>0\ \exists x\in X:\ x>S-\varepsilon)$
4. $\inf(X)\ \Leftrightarrow\ (\forall x\in X: x\ge I)\wedge(\forall\varepsilon>0\ \exists x\in X:\ x<I+\varepsilon)$

**D (Maximum/Minimum):**

• $\max\{x,y\}:=\begin{cases}x, & \text{if } y\le x,\\ y, & \text{if } x\le y.\end{cases}$, $\min\{x,y\}:=\begin{cases}y, & \text{if } y\le x,\\ x, & \text{if } x\le y.\end{cases}$

**T (Properties of the absolute value):**

• Properties:

1. $|x|\ge 0$.
2. $|x+y|\le|x|+|y|$ (triangle inequality).
3. $|x+y|\ge\big||x|-|y|\big|$ (inv. triangle inequality).

**T (Young's inequality):**

• For every $\varepsilon>0$ and all $x,y\in\mathbb{R}$ it holds: $2|xy|\le\varepsilon x^2+\frac{1}{\varepsilon}y^2$.

## 1.2 Vectors and Vector Spaces

**D (inner product):**

• For $x,y\in\mathbb{R}^n$ the standard inner product is $x\cdot y=\langle x,y\rangle=\sum_{i=1}^n x_iy_i$

**D (Euclidean norm):**

• The Euclidean norm is $\|x\|=\sqrt{\sum_{i=1}^n x_i^2}$

**D (Euclidean distance):**

• The Euclidean distance is $d(x,y)=\sqrt{\sum_{i=1}^n(x_i-y_i)^2}$

**D (Triangle inequality):**

• For all $x,y,z\in\mathbb{R}^n$: $\|x-z\|\le\|x-y\|+\|y-z\|$

## 1.3 Complex Numbers

**D (Complex number):**

• A complex number is $z=a+ib,\quad a,b\in\mathbb{R}$

**D (set of complex numbers):**

• The set of complex numbers is $\mathbb{C}=\{a+ib\mid a,b\in\mathbb{R}\}$

**D (Terminology of complex numbers):**

• Terminology

- $a=\mathrm{Re}(z)$ — real part
- $b=\mathrm{Im}(z)$ — imaginary part
- $i^2=-1$
- If $a=0$, the number is purely imaginary

**D (Complex numbers operations):**

• Let $z_1=x_1+iy_1$, $z_2=x_2+iy_2$.

- **addition:** $z_1+z_2=(x_1+x_2)+i(y_1+y_2)$
- **subtraction:** $z_1-z_2=(x_1-x_2)+i(y_1-y_2)$
- **multiplication:** $z_1z_2=x_1x_2-y_1y_2+i(x_1y_2+y_1x_2)$
- **division:** $\frac{z}{w}=\frac{z\overline{w}}{w\overline{w}}=\frac{z\overline{w}}{|w|^2}$
- **conjugate:** $\overline{z}=\overline{x+iy}=x-iy$

**D (Properties of the Conjugate):**

• Properties are:

- $z\overline{z}=|z|^2$
- $z+\overline{z}=2\,\mathrm{Re}(z),\quad z-\overline{z}=2i\,\mathrm{Im}(z)$
- $\overline{\overline{z}}=z$
- $\overline{z\pm w}=\overline{z}\pm\overline{w}$
- $\overline{zw}=\overline{z}\,\overline{w}$
- $z=\overline{z}$, if $z\in\mathbb{R}$
- $\overline{z}=-z$, $z$ is purely imaginary

**D (Modulus):**

• The **magnitude** $|z|$ of a complex number $z=a+ib$ is the length of the corresponding vector: $|z|=\sqrt{a^2+b^2}$ ($\Rightarrow|e^{i\theta}|=\sqrt{\sin(\theta)^2+\cos(\theta)^2}=1$)
• The **distance** between two complex numbers: $d(z_1,z_2)=|z_2-z_1|$
• **Triangle inequality**: $|z+w|\le|z|+|w|$

**D (Euler's Formula):**

• The exponential function can be represented by a **series expansion**

$$e^x=\sum_{k=0}^{\infty}\frac{x^k}{k!}$$

If we substitute $x=it$ into this formula, we obtain **Euler's formula**.

$$e^{it}=\cos t+i\sin t$$

We can **rewrite** $sin(t)$ and $cos(t)$ using complex exponential functions:

$$\cos(t)=\frac{e^{it}+e^{-it}}{2},\quad\sin(t)=\frac{e^{it}-e^{-it}}{2i}$$

**D (Exponential with Complex Argument):**

• We can also allow complex arguments for the exponential function. Then the following rules apply:

- $e^{z+w}=e^ze^w$
- $e^{a+ib}=e^a(\cos b+i\sin b)$
- $e^{z+i2\pi}=e^z$

**D (Polar Form):**

• A complex number can also be described by specifying the distance from the origin and by specifying a suitable angle: $z=re^{i\varphi}$.

- $r=|z|\ge 0$ - distance
- $\varphi\in(-\pi,\pi]$ - polar angle
- $\varphi=\arg(z)=\mathrm{arc}(z)$

**D (Polar to Cartesian conversion):**

• $\boxed{x=r\cos\varphi\mid y=r\sin\varphi}$.

**D (Cartesian to Polar conversion):**

• $\boxed{r=\sqrt{x^2+y^2}\mid\varphi=\arctan\frac{y}{x}}$.

• If $x=0$: $\arg(iy)=\begin{cases}\frac{\pi}{2}, & y>0\\ -\frac{\pi}{2}, & y<0\end{cases}$

**D (Powers of Complex Numbers):**

• The following rules must be observed:

- $z_1z_2=r_1r_2e^{i(\varphi_1+\varphi_2)}$
- $\frac{z_1}{z_2}=\frac{r_1}{r_2}e^{i(\varphi_1-\varphi_2)}$
- $z^n=r^ne^{in\varphi}$

**D (Roots from complex numbers):**

• For $n\in\mathbb{N}$, the solutions to the equation $z^n=re^{i\varphi}$ are

$$z_k=r^{1/n}e^{i(\varphi/n+2\pi k/n)},\quad k=0,\dots,n-1$$

**D (Quadratic Equations):**
• For the quadratic equation $az^2 + bz + c = 0$, in complex numbers the solution is still $z_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$.
**D (Fundamental Theorem of Algebra):**
• Each polynom $p(z) = \sum_{k=0}^{n} a_k z^k, \quad a_k \in \mathbb{C}$ can be factored into linear factors, i.e. written as $p(z) = a_n(z-z_1)(z-z_2)\dots(z-z_n)$ The numbers $z_k$ are therefore precisely the zero points of $p(z)$ (with **multiplicity**).
**C (non-real roots appear in conjugate pairs):**
• Non-real roots of a polynomial with **real** coefficients occur in conjugate pairs $z, \overline{z}$.

## 2 Sequences

**D (Sequence):**
• A sequence $(a_n)_{n\in\mathbb{N}_0}$ (also written $(a_n)_{n\geq 0}$ or $(a_n)_{n=0}^{\infty}$) is an infinite list of numbers, where each entry is called a *term* of the sequence. A sequence can also be viewed as a function $f : \mathbb{N}_0 \to \mathbb{R} \quad (\text{or } f : \mathbb{N} \to \mathbb{R})$, whose values are $f(n) = a_n$.
**D (Ways to Describe a Sequence):**
• One way to describe a sequence is by giving an explicit formula for each term: $a_n = f(n)$. Another possibility is to define the sequence recursively, constructing each new term from previous ones: $a_n = g(a_{n-1}, a_{n-2}, \dots)$. The functions $f$ and $g$ are called the *rule of formation* of the sequence.
**D (Convergence/Divergence):**
• A sequence $(a_n)_{n\in\mathbb{N}_0} \subset \mathbb{R}$ is called *convergent* if there exists a number $L \in \mathbb{R}$ such that

$$\forall \varepsilon > 0 \; \exists N > 0 \text{ such that } \forall n > N : |a_n - L| < \varepsilon.$$

The number $L$ is called the *limit* of the sequence, and we write

$$\lim_{n\to\infty} a_n = L.$$

If no such $L$ exists, the sequence is called *divergent*.
**T (Uniqueness of the Limit):**
• A convergent sequence has exactly one unique limit.
**D (Divergence to infinity):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence.

- If $\forall M > 0 \; \exists N > 0$ such that $\forall n > N : a_n > M$, then the sequence diverges to infinity and we write $\lim_{n\to\infty} a_n = \infty$.
- If $\forall M > 0 \; \exists N > 0$ such that $\forall n > N : a_n < -M$, then the sequence diverges to minus infinity and we write $\lim_{n\to\infty} a_n = -\infty$.

**D (Subsequence):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A *subsequence* is a sequence of the form $(a_{n_k})_{k\in\mathbb{N}_0}$, where $(n_k)_{k\in\mathbb{N}_0}$ is a sequence of non-negative integers satisfying $n_{k+1} > n_k$ for all $k \in \mathbb{N}_0$.
**L (Limit of a sequence and its subsequence):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a convergent sequence with limit $L \in \mathbb{R}$. Then every subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ also converges to the same limit $L$.
**D (Accumulation Point):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is called an accumulation point of the sequence if

$$\forall \varepsilon > 0 \; \forall N \in \mathbb{N}_0 \; \exists n \geq N \text{ such that } |a_n - A| < \varepsilon.$$

Intuitively, every interval $(A-\varepsilon, A+\varepsilon)$ contains terms $a_n$ for infinitely many indices $n$.

**T (Characterization of Accumulation Points):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is an accumulation point of the sequence if and only if there exists a convergent subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ such that $\lim_{k\to\infty} a_{n_k} = A$.
**C (Accumulation Point of a Convergent Sequence):**
• Every convergent sequence $(a_n)_{n\in\mathbb{N}_0}$ in $\mathbb{R}$ has exactly one accumulation point, which coincides with its limit.
**C (Infinite Occurrence Near an Accumulation Point):**
• Let $A$ be an accumulation point of the sequence $(a_n)_{n\in\mathbb{N}_0}$. Then for every $\varepsilon > 0$ there exist infinitely many terms of the sequence lying in the interval $(A-\varepsilon, A+\varepsilon)$.
**T (Algebra of Limits):**
• Assume

$$\lim_{n\to\infty} a_n = K, \qquad \lim_{n\to\infty} b_n = L$$

with $K, L \neq \pm\infty$ and let $C$ be a constant. Then

- $\lim_{n\to\infty}(a_n + b_n) = K + L$
- $\lim_{n\to\infty}(a_n - b_n) = K - L$
- $\lim_{n\to\infty}(a_n b_n) = KL$
- $\lim_{n\to\infty}(Ca_n) = CK$

If $L \neq 0$ and $b_n \neq 0$ then $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{K}{L}$.
If $K < L$ then there exists $N$ such that $a_n < b_n$ for all $n \geq N$.
If there exists $N$ such that $a_n \leq b_n$ for all $n \geq N$, then $K \leq L$.
**T (Important Limits):**
• More on the last page*

- $\lim_{n\to\infty} \frac{\log n}{n} = 0 = \lim_{n\to\infty} \frac{\ln n}{n}$
- $\lim_{n\to\infty} n^{1/n} = 1$
- $\lim_{n\to\infty} x^{1/n} = 1 \qquad (x > 0)$
- $\forall x \in \mathbb{R} : \quad \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\forall x \in \mathbb{R} : \quad \lim_{n\to\infty} \frac{x^n}{n!} = 0$

**T (Sandwich Theorem):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ and $(c_n)_{n\in\mathbb{N}_0}$ be convergent sequences with

$$\lim_{n\to\infty} a_n = L, \qquad \lim_{n\to\infty} c_n = L$$

and suppose there exists $N_0$ such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq N_0.$$

Then $(b_n)_{n\in\mathbb{N}_0}$ is also convergent and $\lim_{n\to\infty} b_n = L$.
**D (Bounded Sequence):**
• A sequence $(a_n)_{n\in\mathbb{N}_0}$ is called bounded below or bounded above if the set $M = \{a_n \mid n \in \mathbb{N}_0\}$ has a lower bound or an upper bound respectively. A sequence that is bounded both above and below is called a bounded sequence.
**D (Monotone Sequence):**
• A sequence $(a_n)_{n\in\mathbb{N}_0}$ is called:

- *monotone decreasing* if $m > n \;\Rightarrow\; a_m \leq a_n$.
- *monotone increasing* if $m > n \;\Rightarrow\; a_m \geq a_n$.
- *strictly monotone decreasing* if $m > n \;\Rightarrow\; a_m < a_n$.
- *strictly monotone increasing* if $m > n \;\Rightarrow\; a_m > a_n$.

**T (Monotone Convergence Theorem):**
• Every bounded and monotone sequence converges.
**L (Convergent sequence = bounded):**
• Every convergent sequence is bounded.
**T (Monotone Convergence):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence of real numbers.

- A monotone sequence converges if and only if it is bounded.
- $(a_n)$ is monoton. increasing $\Rightarrow \lim_{n\to\infty} a_n = \sup\{a_n \mid n \in \mathbb{N}_0\}$.
- $(a_n)$ is monoton. decreasing $\Rightarrow \lim_{n\to\infty} a_n = \inf\{a_n \mid n \in \mathbb{N}_0\}$.

**D (Limit Superior and Limit Inferior):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence.

- *limit superior*: $\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup\{a_k \mid k \geq n\}$
- *limit inferior*: $\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf\{a_k \mid k \geq n\}$

**L (Properties of limsup and liminf):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence. Then:

(i) $\limsup_{n\to\infty} a_n = \inf_{n\in\mathbb{N}} (\sup\{a_k \mid k \geq n\})$
(ii) $\liminf_{n\to\infty} a_n = \sup_{n\in\mathbb{N}} (\inf\{a_k \mid k \geq n\})$
(iii) If $(a_n)$ is convergent, then

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

(iv) A bounded sequence $(a_n)$ converges if and only if

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

**T (Characterization via limsup):**
• Let $(a_n)_{n\in\mathbb{N}_0}$ be a bounded sequence and define $A = \limsup_{n\to\infty} a_n$. Then:

(i) $A$ is an accumulation point of the sequence.
(ii) For every $\varepsilon > 0$, there are only finitely many indices $n$ such that $a_n > A + \varepsilon$.
(iii) For every $\varepsilon > 0$, there are infinitely many indices $n$ such that $A - \varepsilon < a_n < A + \varepsilon$.

An analogous statement holds for the limit inferior.
**C (Subsequence of a Bounded Sequence):**
• Every bounded sequence of real numbers has at least one accumulation point and admits a convergent subsequence.
**D (Cauchy Sequence):**
• A sequence $(a_n)_{n\in\mathbb{N}_0} \subset \mathbb{R}$ is called a *Cauchy sequence* if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : \; |a_n - a_m| < \varepsilon.$$

**L (Cauchy Sequences are Bounded):**
• Every Cauchy sequence is bounded.
**T (Cauchy Criterion):**
• A sequence of real numbers converges if and only if it is a Cauchy sequence.

## 3 Series

**D (Series):**
• A (formal) expression of the form $a_0 + a_1 + a_2 + \cdots = \sum_{n=0}^{\infty} a_n$ is called a *series*. The terms $a_n$ are called elements or summands. If the sequence of partial sums $s_n := \sum_{k=0}^{n} a_k$ converges to a limit $L \in \mathbb{R}$, then the series is said to *converge*, and we write $\sum_{n=0}^{\infty} a_n = L$. Otherwise, the series is called *divergent*.

**T (Necessary Condition for Convergence):**
• If the series $\sum_{n=0}^{\infty} a_n$ converges, then necessarily $\lim_{n\to\infty} a_n = 0$.

**L (Linearity of Series):**
• Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be convergent series. Then:

$$\sum_{n=0}^{\infty}(a_n + b_n) = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n, \qquad \sum_{n=0}^{\infty} C a_n = C \sum_{n=0}^{\infty} a_n \quad (C \in \mathbb{R}).$$

**L (Tail of a Series):**
• Let $\sum_{n=0}^{\infty} a_n$ be a series. For any $N \in \mathbb{N}_0$: $\sum_{n=0}^{\infty} a_n$ converges $\Leftrightarrow$ $\sum_{n=N}^{\infty} a_n$ converges, and in that case

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{N-1} a_n + \sum_{n=N}^{\infty} a_n.$$

**T (Series with Non-Negative Terms):**
• Let $a_n \geq 0$ for all $n$. Then the sequence of partial sums $s_n = \sum_{k=0}^{n} a_k$ is monotonically increasing. If $(s_n)$ is bounded, then the series $\sum_{n=0}^{\infty} a_n$ converges; otherwise, it diverges.

**T (Comparison Test):**
• Let $\sum a_n$, $\sum b_n$, $\sum c_n$ be series with non-negative terms and suppose that for some $N$: $c_n \leq b_n \leq a_n$ for all $n \geq N$. Then:

- If $\sum a_n$ converges, then $\sum b_n$ converges (majorant test).
- If $\sum c_n$ diverges, then $\sum b_n$ diverges (minorant test).

**D (Absolute and Conditional Convergence):**
• A series $\sum a_n$ is called *absolutely convergent* if $\sum |a_n|$ converges.
• It is called *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ does not.

**T (Riemann Rearrangement Theorem):**
• Let $\sum a_n$ be a conditionally convergent series of real numbers, $L \in \mathbb{R}$. Then there exists a bijection $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ such that $\sum_{n=0}^{\infty} a_{\varphi(n)} = L$.

**T (Rearrangement of Absolutely Convergent Series):**
• Let $\sum a_n$ be absolutely convergent and let $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ be a bijection. Then $\sum_{n=0}^{\infty} a_{\varphi(n)}$ converges absolutely and $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} a_{\varphi(n)}$.

**T (Leibniz Criterion):**
• Let $(a_n)$ be a monotonically decreasing sequence of non-negative real numbers with

$$\lim_{n\to\infty} a_n = 0.$$

Then the alternating series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges. Moreover, for all $m \in \mathbb{N}_0$:

$$\sum_{n=0}^{2m+1} (-1)^n a_n \leq \sum_{n=0}^{\infty} (-1)^n a_n \leq \sum_{n=0}^{2m} (-1)^n a_n.$$

**T (Cauchy Criterion for Series):**
• The series $\sum_{n=0}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists $N$ such that for all $n \geq m \geq N$:

$$\left| \sum_{k=m+1}^{n} a_k \right| \leq \varepsilon.$$

**T (Absolute Convergence Implies Convergence):**
• Every absolutely convergent series converges, and

$$\left| \sum_{n=0}^{\infty} a_n \right| \leq \sum_{n=0}^{\infty} |a_n|.$$

**T (Root Test):**
• Let $(a_n)$ be a sequence and define $\rho = \limsup_{n\to\infty} |a_n|^{1/n}$. Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

**T (Ratio Test):**
• Let $(a_n)$ with $a_n \neq 0$ and define $\rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$. Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

**T (Cauchy Product):**
• Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Then

$$\left( \sum_{n=0}^{\infty} a_n \right) \left( \sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} \left( \sum_{k=0}^{n} a_{n-k} b_k \right),$$

and the series on the right converges absolutely.

**D (Power Series):**
• A series of the form $\sum_{k=0}^{\infty} c_k (x-a)^k$ is called a *power series* centered at $a$ with coefficients $c_k$.

**D (Convergence of a Power Series):**
• For fixed $x$, if the sequence of partial sums $s_n(x) = \sum_{k=0}^{n} c_k (x-a)^k$ converges, then the power series is said to converge at $x$.

**T (Radius of Convergence):**
• Every power series has a radius of convergence $R$ such that:

- The series converges for $|x-a| < R$,
- The series diverges for $|x-a| > R$.

The interval $(a-R, a+R)$ is called the interval of convergence. The radius $R$ is given by $\rho = \limsup_{n\to\infty} |c_n|^{1/n}$ , then:

$$R = \begin{cases} 0 & \text{if } \rho = \infty, \\ \frac{1}{\rho} & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

## 4 Functions

### 4.1 Function basics

**D Function:**
• A function (mapping) is a rule that assigns to each admissible input exactly one output in the target set. The set of all possible inputs is called the *domain*, denoted by domain$(f)$ or $D$. The set of all actually attained outputs is called the *image*, denoted by image$(f)$ or $W$.

$$f : \text{domain}(f) \to \text{range}(f), \quad x \mapsto f(x).$$

The input is called the *independent variable*, the output the *dependent variable*.

**D Restriction of a Function:**
• Let $f : D(f) \to \mathbb{R}$ and let $D_0 \subseteq D(f)$. The restriction of $f$ to $D_0$ is the function

$$f|_{D_0} : D_0 \to \mathbb{R}, \quad f|_{D_0}(x) = f(x).$$

Note that $f$ and $f|_{D_0}$ are, in general, different functions.

**D Bounded Functions:**
• Let $f : D(f) \to \mathbb{R}$.

- $f$ is *bounded above* if $\exists M \in \mathbb{R}$ such that $f(x) \leq M \quad \forall x \in D(f)$.
- $f$ is *bounded below* if $\exists M \in \mathbb{R}$ such that $f(x) \geq M \quad \forall x \in D(f)$.
- $f$ is *bounded* if $\exists M \in \mathbb{R}$ such that $|f(x)| \leq M \quad \forall x \in D(f)$.

**D Injective, Surjective, Bijective:**
• Let $f : X \to Y$.

- $f$ is *injective* if (different outputs for different inputs)

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

- $f$ is *surjective* if (every value in the codomain gets hit)

$$\forall y \in Y \; \exists x \in X : f(x) = y.$$

- $f$ is *bijective* if it is both injective and surjective.

**D Composition:**
• Let $f : X \to Y$ and $g : Y \to Z$. The composition is defined by

$$g \circ f : X \to Z, \quad (g \circ f)(x) = g(f(x)).$$

**D Identity and Inverse:**
• For a set $X$, the identity function is defined by $\mathrm{id}_X(x) = x$ $(\forall x \in X)$. If $f : X \to Y$ is bijective, then there exists a unique function $f^{-1} : Y \to X$ such that $f^{-1} \circ f = \mathrm{id}_X$, $f \circ f^{-1} = \mathrm{id}_Y$. This function is called the *inverse* of $f$.

**D Monotonicity:**
• Let $f : D(f) \subseteq \mathbb{R} \to \mathbb{R}$.

- $f$ is *monotonically increasing* if $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$.
- $f$ is *strictly increasing* if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.
- $f$ is *monotonically decreasing* if $x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$.
- $f$ is *strictly decreasing* if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$.

**T Strict Monotonicity Implies Injectivity:**
• Every strictly monotone function is injective.

**D Even and Odd Functions:**
• Let $f : \mathbb{R} \to \mathbb{R}$.

- $f$ is *even* if $f(-x) = f(x)$.
- $f$ is *odd* if $f(-x) = -f(x)$.

**D Convexity and Concavity (Geometric):**
• The graph of a function is called

- *convex* if it bends to the left when moving from left to right,
- *concave* if it bends to the right when moving from left to right.

## 4.2 Limits of Functions

**D Limit of a Function:**
• Let $f : D(f) \to \mathbb{R}$ and $x_0 \in \mathbb{R}$. A number $L \in \mathbb{R}$ is called the limit of $f(x)$ at $x_0$ if

$$\forall \varepsilon > 0\ \exists \delta > 0 \text{ such that } \quad x \in D(f),\ |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The limit is uniquely determined.

**D One-Sided Limits:**
• One-Sided Limits

- Right-hand limit: $\lim_{x \to x_0^+} f(x) = L$
- Left-hand limit: $\lim_{x \to x_0^-} f(x) = L$

**D Limits at Infinity:**
• Limits at Infinity

- $\lim_{x \to \infty} f(x) = L$
- $\lim_{x \to -\infty} f(x) = L$

**D Infinite Limits:**
• Infinite Limits

- $\lim_{x \to c} f(x) = \infty$
- $\lim_{x \to c} f(x) = -\infty$

**T Limit Laws:**
• Let
$$\lim_{x \to c} f(x) = K, \quad \lim_{x \to c} g(x) = L, \quad K, L \neq \pm\infty.$$

Then:

- $\lim_{x \to c}(f + g) = K + L$
- $\lim_{x \to c}(f - g) = K - L$
- $\lim_{x \to c}(fg) = KL$
- $\lim_{x \to c}(Af) = AK$
- If $L \neq 0$: $\lim_{x \to c} \frac{f}{g} = \frac{K}{L}$

## 4.3 Continuity of functions

**D Continuity:**
• Let $f : D \to \mathbb{R}$ and $x_0 \in D$. The function $f$ is called *continuous at* $x_0$ if

$$\forall \varepsilon > 0\ \exists \delta > 0 \text{ such that } |x - x_0| < \delta \ \Rightarrow\ |f(x) - f(x_0)| < \varepsilon.$$

The function is called *continuous* if it is continuous at every point in its domain.

**D One-Sided Continuity:**
• Let $f : D(f) \to \mathbb{R}$ and $x_0 \in D(f)$.

- $f$ is *right-continuous* at $x_0$ if $\lim_{x \to x_0^+} f(x) = f(x_0)$.
- $f$ is *left-continuous* at $x_0$ if $\lim_{x \to x_0^-} f(x) = f(x_0)$.

**T Sequential Characterization of Continuity:**
• Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$ and $x_0 \in D$. Then $f$ is continuous at $x_0$ if and only if, for every sequence $(x_n)_{n \in \mathbb{N}} \subseteq D$,

$$\lim_{n \to \infty} x_n = x_0 \quad \Longrightarrow \quad \lim_{n \to \infty} f(x_n) = f(x_0).$$

**T Algebra of Continuous Functions:**
• Let $f, g : D \subseteq \mathbb{R} \to \mathbb{R}$ be continuous. Then:

- $f + g$ is continuous
- $f - g$ is continuous
- $cf$ is continuous for any constant $c \in \mathbb{R}$
- $f \cdot g$ is continuous
- $\frac{f}{g}$ is continuous if $g \neq 0$

**T Continuity of Composition:**
• Let $f : I \to \text{image}(f)$ and $g : \text{image}(f) \to \mathbb{R}$ be continuous. Then the composition $g \circ f$ is continuous and

$$\lim_{x \to x_0} g(f(x)) = g\left(\lim_{x \to x_0} f(x)\right).$$

**T Continuity of the Inverse:**
• Let $I$ be an interval and let $f : I \to J$ be continuous and bijective. Then $J$ is also an interval and the inverse function $f^{-1} : J \to I$ is continuous.

**T Intermediate Value Theorem:**
• Let $f : [a, b] \to \mathbb{R}$ be continuous and let $c$ be a value between $f(a)$ and $f(b)$. Then there exists $x \in [a, b]$ such that $f(x) = c$.

**C Zero of a Continuous Function (Bolzano theorem):**
• Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous and suppose

$$f(a) < 0, \quad f(b) > 0.$$

Then there exists $x \in (a, b)$ such that

$$f(x) = 0.$$

**D Compact Interval:**
• A bounded and closed interval $[a, b]$ is called *compact*.

**L Bolzano–Weierstrass for Intervals:**
• Let $(x_n)$ be a sequence in a compact interval $[a, b]$. Then there exists a subsequence $(x_{n_k})$ such that $x_{n_k} \to x_0$ with $x_0 \in [a, b]$.

**D Local Extrema:**
• Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$.

- $x_0$ is a *local maximum* if $f(x_0) \geq f(x) \quad \forall x \in D$.
- $x_0$ is a *local minimum* if $f(x_0) \leq f(x) \quad \forall x \in D$.
- Local maxima and minima are called *local extrema*.

**T Extreme Value Theorem:**
• Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous and let $I$ be compact. Then $f$ is bounded and attains its maximum and minimum on $I$, i.e., there exist $x_{\min}, x_{\max} \in I$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ $(\forall x \in I)$.

# 5 Derivatives

## 5.1 Differentiation

**D Derivative:**
• Let $f : D \to \mathbb{R}$ and assume $D$ has no isolated points. The *difference quotient* is

$$\frac{f(a + h) - f(a)}{h}.$$

If the limit exists, the *derivative* of $f$ at $a$ is defined by

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h} = \lim_{b \to a} \frac{f(b) - f(a)}{b - a}.$$

Interpretation:

- instantaneous rate of change
- slope of the tangent
- sign indicates increase/decrease

**D Derivative Function:**
• Given $f : D \to \mathbb{R}$, the *derivative function* is $x \mapsto f'(x)$.

**D Differentiability:**
• A function $f$ is *differentiable at* $x$ if $f'(x)$ exists.If this holds for all $x$ in the domain, $f$ is called differentiable.

**L Differentiability Implies Continuity:**
• If $f$ is differentiable at $x_0$, then $f$ is continuous at $x_0$.

**D One-Sided Derivatives:**
• Derivatives

- Right derivative: $\lim_{h \to 0^+} \frac{f(a+h)-f(a)}{h}$
- Left derivative: $\lim_{h \to 0^-} \frac{f(a+h)-f(a)}{h}$

**Continuity / Differentiability at $x_0$:**
•

$$\boxed{f \text{ continuous at } x_0 \iff \begin{cases} 1.\ f(x_0) \text{ exists}, \\ 2.\ \lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x), \\ 3.\ \lim_{x \to x_0} f(x) = f(x_0). \end{cases}}$$

• **Differentiability:**

$$\boxed{f \text{ differentiable at an interior point } x_0 \iff f'_-(x_0) = f'_+(x_0) \in \mathbb{R}}$$

$$f'_\pm(x_0) := \lim_{h \to 0^\pm} \frac{f(x_0 + h) - f(x_0)}{h}.$$

$$\begin{cases} f'_-(x_0) = f'_+(x_0) \in \mathbb{R} & \Rightarrow f'(x_0) \text{ exists}, \\ f'_-(x_0) \neq f'_+(x_0) & \Rightarrow f'(x_0) \text{ does not exist}, \\ \text{only one one-sided derivative exists} & \Rightarrow f'(x_0) \text{ does not exist}. \end{cases}$$

Example: $f(x) = |x|, \qquad f'_-(0) = -1 \neq 1 = f'_+(0) \Rightarrow f'(0)$ does not exist.

$$\boxed{\begin{aligned} \text{dom}(f) = [a, b]: \qquad & f'_+(a) = \lim_{h \to 0^+} \frac{f(a + h) - f(a)}{h}, \\ & f'_-(b) = \lim_{h \to 0^-} \frac{f(b + h) - f(b)}{h}. \end{aligned}}$$

At an endpoint that belongs to the domain, differentiability is tested using the corresponding one-sided derivative.

• **Mini example:** Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$

**Continuity at** 0:

$$|f(x)| = \left|x^2 \sin\left(\frac{1}{x}\right)\right| \leq x^2 \xrightarrow[x \to 0]{} 0 = f(0).$$

Hence, by the squeeze theorem, $\lim_{x \to 0} f(x) = f(0) = 0$.

**Differentiability at** 0:

$$f'(0) = \lim_{h \to 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \to 0} \frac{h^2 \sin\left(\frac{1}{h}\right)}{h} = \lim_{h \to 0} h \sin\left(\frac{1}{h}\right) = 0,$$

since $\left|h \sin\left(\frac{1}{h}\right)\right| \leq |h| \xrightarrow[h \to 0]{} 0$. Thus $f$ is continuous and differentiable at 0, with

$$\boxed{f'(0) = 0}.$$

**D Higher Derivatives:**
• Define iteratively:

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(n+1)} = (f^{(n)})'.$$

If $f^{(n)}$ exists, $f$ is $n$-times differentiable. If all derivatives up to order $n$ are continuous, then $f \in C^n(D)$. Define $C^\infty(D) := \bigcap_{n=0}^\infty C^n(D)$, called *smooth functions.*

**T Basic Derivatives:**
• Basic Derivatives, more in the end*

- Power: $(ax^p)' = apx^{p-1}$
- Exponential: $(a^x)' = \ln(a)\, a^x$
- Trig.: $(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \frac{1}{\cos^2 x}$
- Hyperbolic: $(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$
- Inverse: $(\ln x)' = \frac{1}{x}, \quad (\arctan x)' = \frac{1}{1+x^2}$

**T Derivative Rules:**
• Derivative Rules

- **Sum and Product Rule.** For differentiable functions $f$ and $g$,

$$(f+g)' = f' + g' \quad (fg)' = f'g + fg'.$$

- **Chain Rule.** If $f$ is differentiable at $x_0$ and $g$ is differentiable at $f(x_0)$, then
$$(g \circ f)'(x_0) = g'\big(f(x_0)\big)\, f'(x_0).$$
- **Quotient Rule.** If $g(x_0) \neq 0$, then

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$

- **Derivative of the Inverse.** Let $f$ be bijective and differentiable, with $f'(x_0) \neq 0$. Then

$$(f^{-1})'\big(f(x_0)\big) = \frac{1}{f'(x_0)}.$$

**T Necessary Condition for Extremum:**
• If $x_0$ is a local extremum and $f$ is differentiable at $x_0$, then $f'(x_0) = 0$.

**T Characterization of Local Extrema:**
• If $x_0$ is a local extremum, then at least one holds:

- $x_0$ is a boundary point
- $f$ is not differentiable at $x_0$
- $f'(x_0) = 0$

**T Rolle's Theorem:**
• Let $f$ be continuous on $[a, b]$ and differentiable on $(a, b)$ with $f(a) = f(b)$.Then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

**T Mean Value Theorem:**
• Let $f$ be continuous on $[a, b]$ and differentiable on $(a, b)$.Then there exists $\xi \in (a, b)$ such that
$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

**T l'Hospital's Rule (0/0):**
• If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ and $\lim_{x\to a} \frac{f'(x)}{g'(x)} = L$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = L.$$

**T l'Hospital's Rule (∞/∞):**
• If $\lim_{x\to a} |f(x)| = \lim_{x\to a} |g(x)| = \infty$ and $\lim_{x\to a} \frac{f'(x)}{g'(x)} = L$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = L.$$

**T l'Hospital at Infinity:**
• If $\lim_{x\to\infty} f(x), g(x) \in \{0, \infty\}$ and $\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = L$, then

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L.$$

**T Monotonicity via Derivative:**
• $f'(x) \geq 0 \iff f$ is increasing

**C Constant Functions:**
• $f$ is constant iff $f'(x) = 0 \quad \forall x$.

**T Convexity via First Derivative:**
• $f$ is convex iff $f'$ is increasing.

**C Convexity via Second Derivative:**
• If $f$ is twice differentiable:

- $f''(x) \geq 0 \iff f$ is convex (curved upward).
- $f''(x) \leq 0 \iff f$ is concave (curved downward)

**T Second Derivative Test:**
• Let $f$ be twice differentiable near $x_0$ and suppose $f'(x_0) = 0$.
Then: $f''(x_0) > 0 \quad \Rightarrow \quad x_0$ is a local minimum,
and $f''(x_0) < 0 \quad \Rightarrow \quad x_0$ is a local maximum.

## 5.2 Taylor Approximation and Power Series

**Local Approximation by Polynomials:**
• In a sufficiently small neighborhood of a point $x_0$, the graph of a differentiable function $f$ behaves almost like its tangent at $x_0$. Hence, the function can locally be approximated by its tangent.

**D Linear Approximation / Tangent Approximation:**
• If $f$ is differentiable at $x_0$, then $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$. The expression $P_1(x) = f(x_0) + f'(x_0)(x - x_0)$ is called the *linearization*, *tangent approximation*, or *first Taylor polynomial* of $f$ at $x_0$.

**T Taylor's Theorem:**
• Let $f$ be infinitely differentiable on an open interval $I$ containing $x_0$. Then for every $x \in I$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n,$$

for some $c$ between $x_0$ and $x$. Equivalently,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_n(x),$$

where $R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c)(x - x_0)^{n+1}$. The polynomial

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

is called the *n-th Taylor polynomial.*

**Taylor Approximation and Power Series:**
• Taylor polynomials provide increasingly accurate approximations of many functions. For many functions, taking infinitely many terms recovers the original function.

**D Taylor Series:**
• For a function $f$, the series $\sum_{k=0}^\infty \frac{f^{(k)}(a)}{k!}(x - a)^k$ is called the *Taylor series* (or Taylor expansion) of $f$ around the point $a$.

**T Important Taylor Series:**
• The following series converge for all $x$:

$$\sin(x) = \sum_{n=0}^\infty (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos(x) = \sum_{n=0}^\infty (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$e^x = \sum_{n=0}^\infty \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

**T Further Important Power Series:**
• The following series converge for $-1 < x < 1$.

$$\ln(1+x) = \sum_{n=1}^\infty (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$\frac{1}{1-x} = \sum_{n=0}^\infty x^n = 1 + x + x^2 + x^3 + \cdots$$

$$(1+x)^p = \sum_{n=0}^\infty \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!}x^2 + \cdots$$

**Operations on Taylor Series:**
• New Taylor series can be constructed from known ones using substitution, multiplication, differentiation, integration.

**T Multiplication of Power Series:**
• Let

$$f(x) = \sum_{n=0}^\infty a_n x^n, \qquad g(x) = \sum_{n=0}^\infty b_n x^n,$$

with $x$ inside both convergence intervals. Then

$$f(x)g(x) = \sum_{n=0}^\infty \left(\sum_{k=0}^n a_k b_{n-k}\right) x^n.$$

**T Termwise Differentiation:**
• Let $f(x) = \sum_{n=0}^\infty a_n x^n$ inside its convergence interval. Then $f$ is differentiable and

$$f'(x) = \sum_{n=1}^\infty n a_n x^{n-1}.$$

**T Termwise Integration:**
• Let $f(x) = \sum_{n=0}^\infty a_n x^n$ inside its convergence interval $I$. Then for every interval $[a, b] \subseteq I$,

$$\int_a^b f(x)\, dx = \sum_{n=0}^\infty \int_a^b a_n x^n\, dx.$$

## 6 Differential equations

**D Differential Equation:**
• A *differential equation* is an equation involving an unknown function together with its derivatives.

$$u'(t) = r(a - u(t))(b - u(t))$$

where $r, a, b$ are constants, another example: $y^{(4)}(x) - 4y^{(3)}(x) + 5y''(x) - 4y'(x) + 4y(x) = 0$. The variables $t, x$ are called *independent variables*, while $u, y$ are called *dependent variables*.

**D Linear Differential Equation:**
• A differential equation of the form

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = s(x)$$

is called a *linear differential equation* with coefficients $a_i(x)$.
• The *order* of the differential equation is the highest order derivative appearing in the equation. The function $s(x)$ is called the *forcing term*.
If $s(x) = 0$, the equation is called *homogeneous*; otherwise it is called *inhomogeneous*.

**D Differential Equation with Constant Coefficients:**
• If all coefficient functions $a_i(x)$ are constant functions, the equation is called a *differential equation with constant coefficients*.

**T Superposition Principle:**
• If $y_1(x)$ and $y_2(x)$ are solutions of a linear homogeneous differential equation, then $y(x) = C_1y_1(x) + C_2y_2(x)$ $(C_1, C_2 \in \mathbb{R})$, is also a solution of the same differential equation.

**T Fundamental Solutions:**
• A linear homogeneous differential equation of order $n$ possesses $n$ solutions $y_1(x), \ldots, y_n(x)$ such that the general solution is given by

$$y(x) = C_1y_1(x) + \cdots + C_ny_n(x), \qquad C_1, \ldots, C_n \in \mathbb{R}.$$

These solutions are called *fundamental solutions*.

**Characteristic Polynomial:**
• Consider the differential equation

$$a_nu^{(n)}(x) + a_{n-1}u^{(n-1)}(x) + \cdots + a_1u'(x) + a_0u(x) = 0.$$

Using the ansatz $u(x) = e^{\lambda x}$ (Euler ansatz), we get the *characteristic polynomial*: $p(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$

**T Solution of Linear Homogeneous Differential Equations:**
• Consider

$$a_nu^{(n)} + \cdots + a_1u' + a_0u = 0.$$

**Step 1: Characteristic equation**

$$a_n\lambda^n + \cdots + a_1\lambda + a_0 = 0.$$

**Step 2: Build one solution block for each root**

| Root of multiplicity $m$ | Contribution to $u_h(x)$ |
|---|---|
| $\lambda \in \mathbb{R}$ | $e^{\lambda x}\left(C_0 + C_1x + \cdots + C_{m-1}x^{m-1}\right)$ |
| $\alpha \pm i\beta$ | $e^{\alpha x}\sum_{k=0}^{m-1} x^k\left(A_k\cos(\beta x) + B_k\sin(\beta x)\right)$ |

**Step 3: Add all blocks**

$$u_h(x) = \text{sum of all root contributions}.$$

**Memory rule:**

$$\boxed{\text{multiplicity } m \Rightarrow 1, x, \ldots, x^{m-1}}$$

$$\boxed{\alpha \pm i\beta \Rightarrow e^{\alpha x}(\cos(\beta x), \sin(\beta x))}$$

**Example: repeated real and complex roots:**
• Suppose $(\lambda - 2)^2(\lambda^2 + 4) = 0$. Roots: $\lambda = 2$ with multiplicity 2, $\lambda = \pm 2i$. Hence $u_h(x) = (C_1 + C_2x)e^{2x} + C_3\cos(2x) + C_4\sin(2x)$.

**Constants in the General Solution:**
• The general solution of a differential equation of order $n$ contains $n$ constants. Therefore, in order to determine a unique solution, one must specify $n$ additional conditions.

**D Boundary Value Problem:**
• Conditions imposed at different points, e.g. for $n = 2$:
$u(t_0) = u_0, \quad u(t_1) = u_1, \quad t_0 \neq t_1,$
define a *boundary value problem*.

**D Initial Value Problem:**
• Conditions imposed at a single point, e.g. for $n = 2$:
$u(t_0) = u_0, \quad u'(t_0) = v_0,$
define an *initial value problem*.

## 7 Integration Part 1

**D Antiderivative / Indefinite Integral:**
• Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a function. A differentiable function $F : I \to \mathbb{R}$ satisfying $F'(x) = f(x) \quad \forall x \in I$ is called an *antiderivative* of $f$ on $I$. Determining $F$ from $f$ is called *indefinite integration*.

**Non-Uniqueness of Antiderivatives:**
• An antiderivative is not unique. If $F(x)$ is an antiderivative of $f(x)$, then $F(x) + C$ is also an antiderivative for every constant $C \in \mathbb{R}$. Hence a function possesses an entire family of antiderivatives. This family is called the *indefinite integral* and is written as

$$\int f(x)\,dx.$$

The function $f$ is called the *integrand*.

**T Linearity of the Indefinite Integral:**
• For indefinite integrals:

$$\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

and

$$\int sf(x)\,dx = s\int f(x)\,dx,$$

where $s$ is a constant.

**T Integration by Parts:**
• Using the product rule: $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$, we obtain $\int \frac{d}{dx}(f(x)g(x))\,dx = (fg)(x)$. Therefore, $(fg)(x) = \int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx$. Rearranging gives the formula for *integration by parts*:

$$\boxed{\int f'(x)g(x)\,dx = f(x)g(x) - \int f(x)g'(x)\,dx}$$

**T Integration by Substitution:**
• Using the chain rule: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$, we obtain $\int f'(g(x))g'(x)\,dx = f(g(x)) + C$. With the substitution $y = g(x)$, this becomes $\int f'(g(x))g'(x)\,dx = \int \frac{df}{dy}\,dy$. Hence the substitution rule is

$$\boxed{\int f'(g(x))g'(x)\,dx = f(g(x)) + C}$$

## 8 Integration Part 2

**D Partition of an Interval:**
• Let $[a, b] \subseteq \mathbb{R}$ be a compact interval. A *partition* of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

The points $x_i$ are called *subdivision points*. A partition

$$a = y_0 < y_1 < \cdots < y_m = b$$

is called a *refinement* of $a = x_0 < x_1 < \cdots < x_n = b$ if $\{x_0, \ldots, x_n\} \subseteq \{y_0, \ldots, y_m\}$.

**D Step Function:**
• A function $f : [a, b] \to \mathbb{R}$ is called a *step function* if there exists a partition $a = x_0 < x_1 < \cdots < x_n = b$ such that for every $k = 1, \ldots, n$, the restriction $f|_{(x_{k-1}, x_k)}$ is constant.

**D Integral of a Step Function:**
• Let $f : [a, b] \to \mathbb{R}$ be a step function with respect to the partition $a = x_0 < x_1 < \cdots < x_n = b$. Then the integral of $f$ over $[a, b]$ is defined by

$$\int_a^b f(x)\,dx := \sum_{k=1}^{n} c_k(x_k - x_{k-1}),$$

where $c_k$ denotes the constant value of $f$ on $(x_{k-1}, x_k)$.

**D Upper and Lower Sums:**
• Let

$$f : [a, b] \to \mathbb{R}.$$

The set of upper sums is defined by

$$U(f) := \left\{ \int_a^b s(x)\,dx \;\middle|\; s \in TF,\ s \geq f \right\},$$

and the set of lower sums by

$$L(f) := \left\{ \int_a^b s(x)\,dx \;\middle|\; s \in TF,\ s \leq f \right\},$$

where $TF$ denotes the set of step functions.

**D Riemann Integral:**
• A bounded function $f : [a, b] \to \mathbb{R}$ is called *Riemann integrable* if

$$\sup L(f) = \inf U(f).$$

In this case, the common value is called the *Riemann integral* and is denoted by

$$\int_a^b f(x)\,dx := \sup L(f) = \inf U(f).$$

The numbers $a, b$ are called the lower and upper limits of integration, and $f$ is called the integrand.

**T Properties of the Riemann Integral:**
• Properties of the Riemann Integral:

(i) (Linearity) If $f, g : [a, b] \to \mathbb{R}$ are integrable and $\alpha, \beta \in \mathbb{R}$, then

$$\int_a^b (\alpha f + \beta g)(x)\,dx = \alpha\int_a^b f(x)\,dx + \beta\int_a^b g(x)\,dx.$$

(ii) (Monotonicity) If $f(x) \leq g(x)$ for all $x$, then

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

(iii) (Triangle Inequality)

$$\left| \int_a^b f(x)\,dx \right| \le \int_a^b |f(x)|\,dx.$$

(iv) (Reversal of Integration Limits)

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

(v) (Splitting the Interval) For $a \le c \le b$:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

**T Monotone Functions are Integrable:**
• Every monotone function $f : [a,b] \to \mathbb{R}$ is Riemann integrable.

**T Continuous Functions are Integrable:**
• Every continuous function $f : [a,b] \to \mathbb{R}$ is Riemann integrable.

**D Pointwise Convergence of Functions:**
• Let $(f_n)_{n\in\mathbb{N}_0}$ be a sequence of functions $f_n : D \to \mathbb{R}$, and let $f : D \to \mathbb{R}$. The sequence $(f_n)$ converges *pointwise* to $f$ if for every $x \in D$, the sequence $(f_n(x))_{n\in\mathbb{N}_0}$ converges to $f(x)$.

**D Uniform Convergence:**
• The sequence $(f_n)$ converges *uniformly* to $f$ if $\forall \varepsilon > 0\ \exists N$ such that for all $n \ge N$ and all $x \in D$, we have $|f_n(x) - f(x)| < \varepsilon$.

**T Uniform Limit of Continuous Functions:**
• Let $(f_n)_{n\in\mathbb{N}_0}$ be a sequence of continuous functions $f_n : D \to \mathbb{R}$ converging uniformly to $f : D \to \mathbb{R}$. Then $f$ is continuous.

**T Uniform Limit and Integration:**
• Let $(f_n)_{n\in\mathbb{N}_0}$ be a sequence of integrable functions $f_n : [a,b] \to \mathbb{R}$ converging uniformly to $f : [a,b] \to \mathbb{R}$. Then $f$ is integrable and

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \int_a^b f_n(x)\,dx.$$

**T Mean Value Theorem for Integrals:**
• If $f : [a,b] \to \mathbb{R}$ is continuous, then there exists $c \in [a,b]$ such that $f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$. The quantity

$$\frac{1}{b-a}\int_a^b f(x)\,dx$$

is called the *average value* of $f$ on $[a,b]$.

**T Fundamental Theorem of Calculus:**
• Let $f : [a,b] \to \mathbb{R}$ be continuous. Then for every constant $C \in \mathbb{R}$, the function

$$F(x) := \int_a^x f(y)\,dy + C$$

is an antiderivative of $f$. Moreover, every antiderivative of $f$ is of this form.

**C Integral Representation:**
• Let $F : [a,b] \to \mathbb{R}$ be continuously differentiable. Then

$$F(x) = F(a) + \int_a^x F'(t)\,dt.$$

**C Newton–Leibniz Formula:**
• Let $f : [a,b] \to \mathbb{R}$ be continuous and let $F$ be an antiderivative of $f$. Then

$$\int_a^b f(t)\,dt = F(b) - F(a).$$

**T Integration by Parts:**
• Let $f, g : [a,b] \to \mathbb{R}$ be continuously differentiable. Then

$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)\,dx.$$

**T Substitution Rule (Version 1):**
• Let $f : I \to J$ be continuously differentiable and $g : J \to \mathbb{R}$ continuous. Then for every $[a,b] \subseteq I$,

$$\int_a^b g(f(x))f'(x)\,dx = \int_{f(a)}^{f(b)} g(y)\,dy.$$

**T Substitution Rule (Version 2):**
• Let $f : I \to J$ be continuously differentiable and $g : J \to \mathbb{R}$ continuous. Assume $f'(x) \neq 0 \quad \forall x \in [a,b]$ and let $f^{-1} : [f(a), f(b)] \to \mathbb{R}$ denote the inverse of $f|_{[a,b]}$. Then

$$\int_a^b g(f(x))\,dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y)\,dy.$$

## 9 Differential equations - 2

**D Separation of Variables:**
• The method of separation of variables is particularly suitable for differential equations of the form $y'(x) = f(x)\cdot g(y)$. The idea is to move all terms involving the independent variable to one side and all terms involving the dependent variable to the other side, and then integrate both sides.

**D Linear Substitution:**
• A differential equation of the form

$$\frac{dy}{dx} = y' = f(ax + by + c), \qquad a, b, c \in \mathbb{R},$$

can be transformed into a differential equation with separable variables by the substitution: $u = ax + by + c$.

**Autonomous Equation:**
• After the substitution $u = ax + by + c$, the differential equation $\frac{dy}{dx} = f(ax + by + c)$ becomes

$$\frac{du}{dx} = a + b\frac{dy}{dx} = a + bf(u).$$

The independent variable $x$ no longer appears explicitly on the right-hand side. Such a differential equation is called *autonomous*.

**D Homogeneous Differential Equation:**
• A differential equation of the form

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right)$$

is called *homogeneous in the variables*. It can be transformed into a differential equation with separable variables by the substitution $u = \frac{y}{x}$.

**D Associated Homogeneous Equation:**
• Given an inhomogeneous differential equation

$$a_n(x)y^{(n)}(x) + \cdots + a_0(x)y(x) = s(x), \qquad s(x) \neq 0,$$

the equation $a_n(x)y^{(n)}(x) + \cdots + a_0(x)y(x) = 0$ is called the associated homogeneous differential equation.

**T General Solution of a Linear Inhomogeneous ODE:**
• Let $y_p(x)$ be a particular solution of a linear inhomogeneous differential equation and let $y_h(x)$ be the general solution of the associated homogeneous equation. Then $y(x) = y_h(x) + y_p(x)$ is the general solution of the linear inhomogeneous differential equation.

**D Variation of Constants:**
• Consider a first-order linear inhomogeneous differential equation. A particular solution can be obtained by replacing the constant appearing in the general solution of the associated homogeneous equation by a function of the independent variable. This method is called *variation of constants*.

**Variation of Constants Example:**
• Given

$$y' - y = x,$$

the associated homogeneous equation

$$y' - y = 0$$

has general solution

$$y_h(x) = Ke^x.$$

The variation-of-constants ansatz is therefore

$$y_p(x) = K(x)e^x.$$

**T Standard Ansätze for Particular Solutions:**
• For a linear inhomogeneous ODE $L(y) = s(t)$:

| Forcing term $s(t)$ | Ansatz $y_p(t)$ |
|---|---|
| $a_0 + a_1t + \cdots + a_nt^n$ | $C_0 + C_1t + \cdots + C_nt^n$ |
| $Ae^{kt}$ | $Ce^{kt}$ |
| $A\sin(\omega t) + B\cos(\omega t)$ | $C_1\sin(\omega t) + C_2\cos(\omega t)$ |

If the corresponding characteristic root has multiplicity $m$, multiply the ansatz by $t^m$. That is: $0 \rightsquigarrow t^m,\ k \rightsquigarrow t^m,\ i\omega \rightsquigarrow t^m$.

**T Picard–Lindelöf–Peano (First Order):**
• The initial value problem

$$y' = f(x, y), \qquad y(x_0) = y_0,$$

has, for sufficiently well-behaved functions $f$, a unique solution $x \mapsto y(x)$, $x \in I$, where the interval $I$ may depend on $x_0$ and $y_0$.

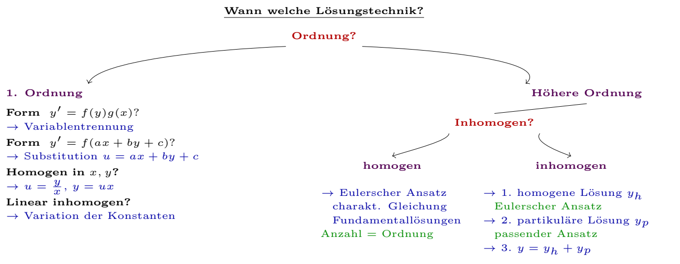
**T Picard–Lindelöf–Peano (Higher Order):**
• The initial value problem

$$y^{(n)} + f_{n-1}(x)y^{(n-1)} + \cdots + f_1(x)y' + f_0(x)y = s(x),$$

with initial conditions

$$y(x_0) = y_0, \qquad y'(x_0) = v_0, \qquad \ldots, \qquad y^{(n-1)}(x_0) = z_0,$$

has, for sufficiently well-behaved functions $f_{n-1}, \ldots, f_0, s$, a unique solution $x \mapsto y(x)$, $x \in I$, where the interval $I$ may depend on $x_0$ and the prescribed initial data.



### Important Limits

- $\lim_{x\to\infty} \frac{1}{x} = 0$
- $\lim_{x\to\infty} \left(1+\frac{1}{x}\right) = 1$
- $\lim_{x\to 0} \frac{\log(1-x)}{x} = -1$
- $\lim_{x\to 0} x\log x = 0$
- $\lim_{x\to\infty} e^x = \infty$
- $\lim_{x\to-\infty} e^x = 0$
- $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
- $\lim_{x\to 0} \frac{e^x-1}{x} = 1$
- $\lim_{x\to\infty} e^{-x} = 0$
- $\lim_{x\to-\infty} e^{-x} = \infty$
- $\lim_{x\to 0} \frac{x}{\arctan x} = 1$
- $\lim_{x\to\infty} \arctan x = \frac{\pi}{2}$
- $\lim_{x\to\infty} \frac{e^x}{x^m} = \infty$
- $\lim_{x\to-\infty} xe^x = 0$
- $\lim_{x\to\infty} \left(\frac{x}{x+k}\right)^x = e^{-k}$
- $\lim_{x\to\infty} \ln x = \infty$
- $\lim_{x\to 0} \ln x = -\infty$
- $\lim_{x\to 0} \frac{a^x-1}{x} = \ln(a)$
- $\lim_{x\to 0} \frac{e^{ax}-1}{x} = a$
- $\lim_{x\to\infty}(1+x)^{1/x} = 1$
- $\lim_{x\to 0}(1+x)^{1/x} = e$
- $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x\to 1} \frac{\ln x}{x-1} = 1$
- $\lim_{x\to\infty} \frac{\ln x}{x} = 0$
- $\lim_{x\to\infty} \frac{\log x}{x^\alpha} = 0$
- $\lim_{x\to\infty} x^a q^x = 0,\ 0\le q<1$
- $\lim_{x\to\infty} n^{1/n} = 1$
- $\lim_{x\to\infty} \sqrt[x]{x} = 1$
- $\lim_{x\to\infty} \frac{2x}{2^x} = 0$
- $\lim_{x\to\pm\infty} \left(1+\frac{1}{x}\right)^x = e$
- $\lim_{x\to\infty} \left(1-\frac{1}{x}\right)^x = \frac{1}{e}$
- $\lim_{x\to\pm\infty} \left(1+\frac{k}{x}\right)^{mx} = e^{km}$
- $\lim_{x\to 0} \frac{\sin x}{x} = 1$
- $\lim_{x\to\infty} \frac{\sin x}{x} = 0$
- $\lim_{x\to 0^+} x\ln x = 0$
- $\lim_{x\to(\pi/2)^-} \tan x = +\infty$
- $\lim_{x\to(\pi/2)^+} \tan x = -\infty$
- $\lim_{x\to 0} \frac{1}{\cos x} = 1$
- $\lim_{x\to 0} \frac{\cos x - 1}{x} = 0$
- $\lim_{x\to 0} \sqrt[n]{n!} = \infty$
- $\lim_{x\to\infty}(\log n)^{1/n} = 1$

### Rules of derivation (holds only if $f^{-1}$ exists and $f$ continuous.)

- **Sum:** $(f+g)' = f'+g'$
- **Product:** $(fg)' = f'g+fg'$
- **Chain rule:** $(g\circ f)' = g'(f)f'$
- **Quotient rule:** $\left(\frac{f}{g}\right)' = \frac{f'g-fg'}{g^2}$
- **Derivative of the Inverse:** $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \quad y_0 = f(x_0)$

### Rules of integration

- **Linearity of Integration:** $\int(f+g) = \int f + \int g$ and $\int cf = c\int f$
- **Integration by Parts:** $\int f'g = fg - \int fg'$
- **Substitution:** $\int g(f(x))f'(x)\,dx = \int g(y)\,dy$

### Logarithms

- **Change of base:** $\log_a x = \frac{\ln x}{\ln a}$
- **Power rule:** $\log_a(x^y) = y\log_a x$
- **Product rule:** $\log_a(xy) = \log_a x + \log_a y$
- **Quotient rule:** $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
- **Logarithm 1:** $\log_a(1) = 0\ \forall a\in\mathbb{N}$ ($x\le 0$ undefined)

### Trigonometry

- **Tangent and cotangent:** $\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$
- **Pythagorean identity:** $\sin^2 x + \cos^2 x = 1$
- **Double-angle sine:** $\sin(2x) = 2\sin x\cos x$
- **Double-angle cosine:** $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$
- **Angle addition sine:** $\sin(x+y) = \sin x\cos y + \cos x\sin y$
- **Angle addition cosine:** $\cos(x+y) = \cos x\cos y - \sin x\sin y$
- **Even/Odd symmetry:** $\cos(x) = \cos(-x)$ and $\sin(-x) = -\sin(x)$
- **Reflection at $\pi/2$:** $\cos(\pi-x) = -\cos(x)$ and $\sin(\pi-x) = \sin(x)$

### Special Trigonometric Values

| $^\circ$ | rad | $\sin(\xi)$ | $\cos(\xi)$ | $\tan(\xi)$ |
|---|---|---|---|---|
| $0^\circ$ | $0$ | $0$ | $1$ | $0$ |
| $30^\circ$ | $\frac{\pi}{6}$ | $\frac{1}{2}$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{3}}{3}$ |
| $45^\circ$ | $\frac{\pi}{4}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{2}}{2}$ | $1$ |
| $60^\circ$ | $\frac{\pi}{3}$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{2}$ | $\sqrt{3}$ |
| $90^\circ$ | $\frac{\pi}{2}$ | $1$ | $0$ | $\varnothing$ |
| $120^\circ$ | $\frac{2\pi}{3}$ | $\frac{\sqrt{3}}{2}$ | $-\frac{1}{2}$ | $-\sqrt{3}$ |
| $135^\circ$ | $\frac{3\pi}{4}$ | $\frac{\sqrt{2}}{2}$ | $-\frac{\sqrt{2}}{2}$ | $-1$ |
| $150^\circ$ | $\frac{5\pi}{6}$ | $\frac{1}{2}$ | $-\frac{\sqrt{3}}{2}$ | $-\frac{\sqrt{3}}{3}$ |
| $180^\circ$ | $\pi$ | $0$ | $-1$ | $0$ |

| Quadrant | Angle range | sin | cos | tan |
|---|---|---|---|---|
| I | $0^\circ$ to $90^\circ$ | + | + | + |
| II | $90^\circ$ to $180^\circ$ | + | − | − |
| III | $180^\circ$ to $270^\circ$ | − | − | + |
| IV | $270^\circ$ to $360^\circ$ | − | + | − |

### Hyperbolic

- **Hyperbolic sine:** $\sinh(x) := \frac{e^x-e^{-x}}{2} : \mathbb{R}\to\mathbb{R}$
- **Hyperbolic cosine:** $\cosh(x) := \frac{e^x+e^{-x}}{2} : \mathbb{R}\to[1,\infty)$
- **Hyperbolic tangent:** $\tanh(x) := \frac{\sinh(x)}{\cosh(x)} = \frac{e^x-e^{-x}}{e^x+e^{-x}} : \mathbb{R}\to(-1,1)$
- **Inverse hyperbolic sine:** $\operatorname{arsinh} x = \ln(x+\sqrt{x^2+1})$
- **Inverse hyperbolic cosine:** $\operatorname{arcosh} x = \ln(x+\sqrt{x^2-1})$
- **Inverse hyperbolic tangent:** $\operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$

### Complement trick

$$\sqrt{ax+b}-\sqrt{cx+d} = \frac{ax+b-(cx+d)}{\sqrt{ax+b}+\sqrt{cx+d}}$$

### Derivatives and Antiderivatives

| **Antiderivative** | **Function** | **Derivative** |
|---|---|---|
| $\frac{x^{n+1}}{n+1}$ | $x^n$ | $nx^{n-1}$ |
| $\ln\lvert x\rvert$ | $\frac{1}{x} = x^{-1}$ | $-x^{-2} = -\frac{1}{x^2}$ |
| $\frac{2}{3}x^{3/2}$ | $\sqrt{x} = x^{1/2}$ | $\frac{1}{2\sqrt{x}}$ |
| $\frac{n}{n+1}x^{1/n+1}$ | $\sqrt[n]{x} = x^{1/n}$ | $\frac{1}{n}x^{1/n-1}$ |
| $e^x$ | $e^x$ | $e^x$ |
| $\exp(x)$ | $\exp(x)$ | $\exp(x)$ |
| $\frac{1}{a(n+1)}(ax+b)^{n+1}$ | $(ax+b)^n$ | $n(ax+b)^{n-1}a$ |
| $x(\ln\lvert x\rvert-1)$ | $\ln(x)$ | $\frac{1}{x}$ |
| $\frac{1}{\ln(a)}a^x$ | $a^x$ | $a^x\ln(a)$ |
| $\frac{x}{\ln(a)}(\ln\lvert x\rvert-1)$ | $\log_a\lvert x\rvert$ | $\frac{1}{x\ln(a)}$ |
| $-\cos(x)$ | $\sin(x)$ | $\cos(x)$ |
| $\sin(x)$ | $\cos(x)$ | $-\sin(x)$ |
| $-\ln\lvert\cos(x)\rvert$ | $\tan(x)$ | $\frac{1}{\cos^2(x)}$ |
| $\ln\lvert\sin(x)\rvert$ | $\cot(x)$ | $-\frac{1}{\sin^2(x)}$ |
| $x\arcsin(x)+\sqrt{1-x^2}$ | $\arcsin(x)$ | $\frac{1}{\sqrt{1-x^2}}$ |
| $x\arccos(x)-\sqrt{1-x^2}$ | $\arccos(x)$ | $-\frac{1}{\sqrt{1-x^2}}$ |
| $x\arctan(x)-\frac{1}{2}\ln(x^2+1)$ | $\arctan(x)$ | $\frac{1}{x^2+1}$ |
| $\ln\lvert\sin(x)\rvert$ | $\cot(x)$ | $-\frac{1}{\sin^2(x)}$ |
| $\cosh(x)$ | $\sinh(x)$ | $\cosh(x)$ |
| $\sinh(x)$ | $\cosh(x)$ | $\sinh(x)$ |
| $\ln\lvert\cosh(x)\rvert$ | $\tanh(x)$ | $\frac{1}{\cosh^2(x)}$ |

| $F(x)$ (**further derivatives** $\Rightarrow$) | $f(x) = F'(x)$ |
|---|---|
| $\frac{1}{a}\ln\lvert ax+b\rvert$ | $\frac{1}{ax+b}$ |
| $\frac{ax}{c}-\frac{ad-bc}{c^2}\ln\lvert cx+d\rvert$ | $\frac{a(cx+d)-c(ax+b)}{(cx+d)^2}$ |
| $\frac{x}{2}\sqrt{a^2+x^2}+\frac{a^2}{2}\ln\lvert x+\sqrt{a^2+x^2}\rvert$ | $\sqrt{a^2+x^2}$ |
| $\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\arcsin\left(\frac{x}{\lvert a\rvert}\right)$ | $\sqrt{a^2-x^2}$ |
| $\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\ln\lvert x+\sqrt{x^2-a^2}\rvert$ | $\sqrt{x^2-a^2}$ |
| $\ln\lvert x+\sqrt{x^2\pm a^2}\rvert$ | $\frac{1}{\sqrt{x^2\pm a^2}}$ |
| $\arcsin\left(\frac{x}{\lvert a\rvert}\right)$ | $\frac{1}{\sqrt{a^2-x^2}}$ |
| $\frac{1}{a}\arctan\left(\frac{x}{a}\right)$ | $\frac{1}{a^2+x^2}$ |
| $-\frac{1}{a}\cos(ax+b)$ | $\sin(ax+b)$ |
| $\frac{1}{a}\sin(ax+b)$ | $\cos(ax+b)$ |
| $x^x$ | $x^x(1+\ln\lvert x\rvert)$ |
| $(x^x)^x$ | $(x^x)^x(x+2x\ln\lvert x\rvert)$ |
| $x^{x^x}$ | $x^{x^x}\left(x^{x-1}+x^x(1+\ln\lvert x\rvert)\ln\lvert x\rvert\right)$ |
| $\frac{1}{2}\left(x-\frac{1}{2}\sin(2x)\right)$ | $\sin^2(x)$ |
| $\frac{1}{2}\left(x+\frac{1}{2}\sin(2x)\right)$ | $\cos^2(x)$ |
| $\operatorname{arsinh}(x)$ | $\frac{1}{\sqrt{1+x^2}}$ |
| $\operatorname{arcosh}(x)$ | $\frac{1}{\sqrt{x^2-1}}$ |
| $\operatorname{artanh}(x)$ | $\frac{1}{1-x^2}$ |

## Essential Implications

| | |
|---|---|
| **Regularity** | $C^\infty \Rightarrow C^n \Rightarrow C^1 \Rightarrow$ differentiable $\Rightarrow$ continuous |
| **Derivative bound** | $\lvert f'\rvert \le M \Rightarrow$ Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous |
| **Derivative sign** | $f' > 0 \Rightarrow f$ strictly increasing $\Rightarrow f$ injective, $\quad f' < 0 \Rightarrow f$ strictly decreasing $\Rightarrow f$ injective |
| **Monotonicity** | $f' \ge 0 \Rightarrow f$ increasing, $\quad f' \le 0 \Rightarrow f$ decreasing |
| **Constant function** | $f' = 0 \iff f =$ constant, $\quad f' = g' \Rightarrow f - g = C$ |
| **Inverse** | $f$ strictly monotone $\Rightarrow f$ injective $\Rightarrow f^{-1}$ exists on $f(D)$ |
| **Continuous inverse** | $f : I \to J$ continuous and bijective $\Rightarrow f^{-1} : J \to I$ continuous |
| **Extremum condition** | $x_0$ interior local extremum + $f$ differentiable at $x_0 \Rightarrow f'(x_0) = 0$ |
| **First derivative test** | $f' : - \to + \Rightarrow x_0$ local minimum, $\quad f' : + \to - \Rightarrow x_0$ local maximum |
| **Second derivative test** | $f'(x_0) = 0,\ f''(x_0) > 0 \Rightarrow x_0$ local minimum, $\quad f'(x_0) = 0,\ f''(x_0) < 0 \Rightarrow x_0$ local maximum |
| **Convexity** | $f'' \ge 0 \Rightarrow f'$ increasing $\Rightarrow f$ convex, $\quad f'' \le 0 \Rightarrow f$ concave |
| **Convex minimum** | $f$ convex, $f'(x_0) = 0 \Rightarrow x_0$ global minimum |
| **Continuous compact** | $f \in C([a,b]) \Rightarrow f$ bounded and attains $\min, \max$ |
| **Intermediate value** | $f \in C([a,b]),\ f(a)f(b) < 0 \Rightarrow \exists c \in (a,b) : f(c) = 0$ |
| **Unique root** | continuity + sign change + strict monotonicity $\Rightarrow \exists! c : f(c) = 0$ |
| **Sequential continuity** | $f$ continuous at $x_0 \iff (x_n \to x_0 \Rightarrow f(x_n) \to f(x_0))$ |
| **Sequence convergence** | $a_n \to L \Rightarrow (a_n)$ bounded |
| **Subsequences** | $a_n \to L \Rightarrow a_{n_k} \to L$ for every subsequence |
| **Divergence via subsequences** | $a_{n_k} \to A,\ a_{m_k} \to B,\ A \ne B \Rightarrow (a_n)$ diverges |
| **Accumulation point** | $A$ accumulation point $\iff \exists (a_{n_k}) : a_{n_k} \to A$ |
| **Bolzano–Weierstrass** | $(a_n)$ bounded $\Rightarrow \exists$ convergent subsequence |
| **Monotone convergence** | monotone + bounded $\Rightarrow$ convergent |
| **Increasing sequence** | $a_n \uparrow,\ (a_n)$ bounded above $\Rightarrow a_n \to \sup\{a_n : n \in \mathbb{N}\}$ |
| **Decreasing sequence** | $a_n \downarrow,\ (a_n)$ bounded below $\Rightarrow a_n \to \inf\{a_n : n \in \mathbb{N}\}$ |
| **Cauchy** | in $\mathbb{R}$: $\quad (a_n)$ convergent $\iff (a_n)$ Cauchy $\Rightarrow (a_n)$ bounded |
| **Order and limits** | $a_n \le b_n$ eventually, $a_n \to A,\ b_n \to B \Rightarrow A \le B$ |
| **Sandwich theorem** | $a_n \le b_n \le c_n,\ a_n \to L,\ c_n \to L \Rightarrow b_n \to L$ |
| **Useful squeeze form** | $\lvert b_n\rvert \le c_n,\ c_n \to 0 \Rightarrow b_n \to 0$ |
| **Limsup/liminf** | $(a_n)$ bounded: $\quad a_n$ converges $\iff \liminf a_n = \limsup a_n$ |
| **Series term test** | $\sum a_n$ converges $\Rightarrow a_n \to 0$ |
| **Series divergence** | $a_n \not\to 0 \Rightarrow \sum a_n$ diverges |
| **Absolute convergence** | $\sum \lvert a_n\rvert$ converges $\Rightarrow \sum a_n$ converges $\Rightarrow a_n \to 0$ |
| **Comparison test** | $0 \le b_n \le a_n,\ \sum a_n$ converges $\Rightarrow \sum b_n$ converges |
| **Divergence comparison** | $0 \le b_n \le a_n,\ \sum b_n$ diverges $\Rightarrow \sum a_n$ diverges |
| **Ratio test** | $\lim \left\lvert \frac{a_{n+1}}{a_n} \right\rvert < 1 \Rightarrow \sum a_n$ absolutely convergent, $\quad > 1 \Rightarrow$ divergent |
| **Root test** | $\limsup \lvert a_n\rvert^{1/n} < 1 \Rightarrow \sum a_n$ absolutely convergent, $\quad > 1 \Rightarrow$ divergent |
| **Power series** | $\lvert x - a\rvert < R \Rightarrow$ absolute convergence, $\quad \lvert x - a\rvert > R \Rightarrow$ divergence, $\lvert x - a\rvert = R \Rightarrow$ check separately |
| **Inside radius** | $\lvert x - a\rvert < R \Rightarrow$ termwise differentiation/integration $\Rightarrow C^\infty$ |
| **Riemann integrability** | $f$ continuous on $[a,b] \Rightarrow f$ Riemann integrable $\Rightarrow f$ bounded |
| **Monotone integrability** | $f$ monotone on $[a,b] \Rightarrow f$ Riemann integrable |
| **Integral order** | $f \le g \Rightarrow \int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$ |
| **Integral bound** | $\lvert f\rvert \le M \Rightarrow \left\lvert \int_a^b f(x)\,dx \right\rvert \le M\lvert b - a\rvert$ |
| **Antiderivatives** | $F' = f,\ G' = f \Rightarrow F - G = C$ |
| **FTC I** | $f \in C([a,b]),\ F(x) = \int_a^x f(t)\,dt \Rightarrow F' = f$ |
| **FTC II** | $F' = f \Rightarrow \int_a^b f(x)\,dx = F(b) - F(a)$ |
| **Uniform convergence** | $f_n \xrightarrow{\text{unif}} f \Rightarrow f_n \xrightarrow{\text{pointwise}} f$ |
| **Uniform + continuity** | $f_n$ continuous, $f_n \xrightarrow{\text{unif}} f \Rightarrow f$ continuous |
| **Uniform + integration** | $f_n$ integrable, $f_n \xrightarrow{\text{unif}} f \Rightarrow \lim_{n\to\infty} \int_a^b f_n = \int_a^b f$ |

| | |
|---|---|
| **False converse** | continuous $\not\Rightarrow$ differentiable ($\lvert x\rvert$) |
| **False converse** | bounded sequence $\not\Rightarrow$ convergent ($(-1)^n$) |
| **False converse** | $a_n \to 0 \not\Rightarrow \sum a_n$ convergent $\left(\sum \frac{1}{n}\right)$ |
| **False converse** | $f'(x_0) = 0 \not\Rightarrow x_0$ extremum ($x^3$) |
| **False converse** | $f$ strictly increasing $\not\Rightarrow f' > 0$ ($x^3$) |
| **False converse** | pointwise $\not\Rightarrow$ uniform |
| **False converse** | Riemann integrable $\not\Rightarrow$ continuous |
| **False converse** | convergent series $\not\Rightarrow$ absolutely convergent |
| **Regularity** | $C^\infty \Rightarrow C^n \Rightarrow C^1 \Rightarrow$ differentiable $\Rightarrow$ continuous |

| | |
|---|---|
| **Continuity of composition (proof)** | Let $f$ be continuous at $x_0$ and $g$ continuous at $f(x_0)$. Given $\varepsilon > 0$, choose $\eta > 0$ such that $\lvert y - f(x_0)\rvert < \eta \Rightarrow \lvert g(y) - g(f(x_0))\rvert < \varepsilon$. Choose $\delta > 0$ such that $\lvert x - x_0\rvert < \delta \Rightarrow \lvert f(x) - f(x_0)\rvert < \eta$. Then $\lvert x - x_0\rvert < \delta \Rightarrow \lvert g(f(x)) - g(f(x_0))\rvert < \varepsilon$. |

## Essential Proofs I

| | |
|---|---|
| **Uniqueness of the limit** | Assume $a_n \to A$ and $a_n \to B$. For arbitrary $\varepsilon > 0$, choose $N_A, N_B$ such that $n \ge N_A \Rightarrow \lvert a_n - A\rvert < \varepsilon/2$ and $n \ge N_B \Rightarrow \lvert a_n - B\rvert < \varepsilon/2$. For $n \ge N := \max\{N_A, N_B\}$, $\lvert A - B\rvert \le \lvert A - a_n\rvert + \lvert a_n - B\rvert < \varepsilon$. Since this holds for every $\varepsilon > 0$, $\lvert A - B\rvert = 0$, hence $A = B$. |
| **Subsequences preserve limits** | Assume $a_n \to L$ and let $(a_{n_k})$ be a subsequence. For arbitrary $\varepsilon > 0$, choose $N$ such that $n \ge N \Rightarrow \lvert a_n - L\rvert < \varepsilon$. Since $(n_k)$ is strictly increasing, $n_k \ge k$. Thus, for $k \ge N$, $n_k \ge k \ge N \Rightarrow \lvert a_{n_k} - L\rvert < \varepsilon$. Therefore $a_{n_k} \to L$. |
| **Accumulation point gives infinitely many terms** | Let $A$ be an accumulation point. Then there exists a subsequence $a_{n_k} \to A$. For arbitrary $\varepsilon > 0$, choose $K$ such that $k \ge K \Rightarrow \lvert a_{n_k} - A\rvert < \varepsilon$. Hence $a_{n_k} \in (A - \varepsilon, A + \varepsilon)$ for every $k \ge K$. Since the indices $n_k$ are distinct and there are infinitely many $k \ge K$, the interval contains infinitely many sequence terms. |
| **Convergent sequences are bounded** | Assume $a_n \to L$. Choose $N$ such that $n \ge N \Rightarrow \lvert a_n - L\rvert < 1$. For $n \ge N$, $\lvert a_n\rvert \le \lvert a_n - L\rvert + \lvert L\rvert < 1 + \lvert L\rvert$. Define $M := \max\{\lvert a_0\rvert, \ldots, \lvert a_{N-1}\rvert, 1 + \lvert L\rvert\}$. Then $\lvert a_n\rvert \le M$ for every $n \in \mathbb{N}_0$; hence $(a_n)$ is bounded. |
| **Sandwich theorem** | Assume $a_n \to L$, $c_n \to L$, and $a_n \le b_n \le c_n$ for all $n \ge N_0$. For arbitrary $\varepsilon > 0$, choose $N_1, N_2$ such that $n \ge N_1 \Rightarrow L - \varepsilon < a_n < L + \varepsilon$ and $n \ge N_2 \Rightarrow L - \varepsilon < c_n < L + \varepsilon$. For $n \ge N := \max\{N_0, N_1, N_2\}$, $L - \varepsilon < a_n \le b_n \le c_n < L + \varepsilon$. Thus $\lvert b_n - L\rvert < \varepsilon$, so $b_n \to L$. |
| **Useful squeeze form** | If $\lvert b_n\rvert \le c_n$ eventually and $c_n \to 0$, then $-c_n \le b_n \le c_n$. Since $-c_n \to 0$ and $c_n \to 0$, the sandwich theorem gives $b_n \to 0$. |
| **Order preservation under limits** | Assume $a_n \le b_n$ eventually, $a_n \to A$, and $b_n \to B$. Suppose $A > B$ and set $\varepsilon := (A - B)/3 > 0$. Eventually, $a_n > A - \varepsilon$ and $b_n < B + \varepsilon$. But $A - \varepsilon > B + \varepsilon$, so eventually $a_n > b_n$, contradicting $a_n \le b_n$. Hence $A \le B$. |
| **Unique accumulation point of a convergent sequence** | If $a_n \to L$, every convergent subsequence also converges to $L$. Since every accumulation point is the limit of some subsequence, every accumulation point equals $L$. Moreover, $(a_n)$ itself converges to $L$, so $L$ is an accumulation point. Hence $L$ is the unique accumulation point. |
| **Nonnegative series** | Let $s_n := \sum_{k=0}^n a_k$ with $a_k \ge 0$. Then $s_{n+1} - s_n = a_{n+1} \ge 0$, so $(s_n)$ is increasing. Hence $(s_n)$ bounded $\Rightarrow (s_n)$ convergent $\Rightarrow \sum_{n=0}^\infty a_n$ converges. If $(s_n)$ is unbounded, then the series diverges. |
| **Necessary condition for series convergence** | Assume $\sum_{n=0}^\infty a_n$ converges and let $s_n := \sum_{k=0}^n a_k \to S$. Since $a_n = s_n - s_{n-1}$, we obtain $\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = S - S = 0$. Thus $\sum a_n$ convergent $\Rightarrow a_n \to 0$. |
| **Cauchy criterion: convergence $\Rightarrow$ Cauchy** | Assume $a_n \to L$. For arbitrary $\varepsilon > 0$, choose $N$ such that $n \ge N \Rightarrow \lvert a_n - L\rvert < \varepsilon/2$. Then for all $m, n \ge N$, $\lvert a_n - a_m\rvert \le \lvert a_n - L\rvert + \lvert a_m - L\rvert < \varepsilon$. Hence $(a_n)$ is Cauchy. |
| **Cauchy criterion: Cauchy $\Rightarrow$ convergence** | Let $(a_n)$ be Cauchy. Then $(a_n)$ is bounded, so by Bolzano–Weierstrass there exists a subsequence $a_{n_k} \to L$. For $\varepsilon > 0$, choose $N_1$ such that $m, n \ge N_1 \Rightarrow \lvert a_n - a_m\rvert < \varepsilon/2$, and choose $K$ such that $k \ge K \Rightarrow \lvert a_{n_k} - L\rvert < \varepsilon/2$. Choose $k$ with $k \ge K$ and $n_k \ge N_1$. Then for every $n \ge N_1$, $\lvert a_n - L\rvert \le \lvert a_n - a_{n_k}\rvert + \lvert a_{n_k} - L\rvert < \varepsilon$. Thus $a_n \to L$. |
| **Cauchy sequences are bounded** | Let $(a_n)$ be Cauchy. Choose $N$ such that $m, n \ge N \Rightarrow \lvert a_n - a_m\rvert < 1$. Fix $m = N$. Then for $n \ge N$, $\lvert a_n\rvert \le \lvert a_n - a_N\rvert + \lvert a_N\rvert < 1 + \lvert a_N\rvert$. Set $M := \max\{\lvert a_0\rvert, \ldots, \lvert a_{N-1}\rvert, 1 + \lvert a_N\rvert\}$. Then $\lvert a_n\rvert \le M$ for every $n$, so $(a_n)$ is bounded. |
| **Monotone sequence with bounded subsequence** | Assume $(a_n)$ is increasing and has a bounded subsequence $\lvert a_{n_k}\rvert \le M$. For every $n$, choose $k$ with $n \le n_k$. By monotonicity, $a_n \le a_{n_k} \le M$. Thus $(a_n)$ is bounded above. Since it is increasing and bounded, it converges. The decreasing case is analogous. |
| **Differentiable $\Rightarrow$ continuous** | Assume $f$ is differentiable at $x_0$. For $h \ne 0$, $f(x_0 + h) - f(x_0) = \frac{f(x_0+h) - f(x_0)}{h}\,h$. Hence $\lim_{h\to 0}(f(x_0 + h) - f(x_0)) = f'(x_0) \cdot 0 = 0$. Thus $\lim_{x\to x_0} f(x) = f(x_0)$, so $f$ is continuous at $x_0$. |
| **Rolle's theorem** | Since $f \in C([a,b])$, it attains a minimum and maximum. If $f$ is constant, then $f' = 0$ on $(a,b)$. Otherwise, because $f(a) = f(b)$, at least one extremum occurs at some $\xi \in (a,b)$. Since $f$ is differentiable there, Fermat's theorem gives $f'(\xi) = 0$. |
| **Mean value theorem** | Define $g(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$. Then $g$ is continuous on $[a,b]$, differentiable on $(a,b)$, and $g(a) = g(b) = f(a)$. By Rolle, some $\xi \in (a,b)$ satisfies $0 = g'(\xi) = f'(\xi) - \frac{f(b) - f(a)}{b - a}$. Hence $f'(\xi) = \frac{f(b) - f(a)}{b - a}$. |
| **Strict monotonicity $\Rightarrow$ injectivity** | Let $x_1 \ne x_2$; w.l.o.g. $x_1 < x_2$. If $f$ is strictly increasing, then $f(x_1) < f(x_2)$; if strictly decreasing, then $f(x_1) > f(x_2)$. Thus $f(x_1) \ne f(x_2)$, so $f$ is injective. |
| **Algebra of continuous functions** | Let $x_n \to x_0$. Continuity gives $f(x_n) \to f(x_0)$ and $g(x_n) \to g(x_0)$. By the limit laws, $f(x_n) \pm g(x_n) \to f(x_0) \pm g(x_0)$, $f(x_n)g(x_n) \to f(x_0)g(x_0)$, and, if $g(x_0) \ne 0$, $f(x_n)/g(x_n) \to f(x_0)/g(x_0)$. Hence the corresponding functions are continuous. |

# Essential Proofs II

**Bolzano–Weierstrass on $[a, b]$**
A sequence $(x_n) \subseteq [a, b]$ is bounded. By Bolzano–Weierstrass, it has a convergent subsequence $x_{n_k} \to x_0$. Since $a \le x_{n_k} \le b$ for all $k$, order preservation under limits yields $a \le x_0 \le b$. Thus $x_0 \in [a, b]$.

**Continuous on compact $\Rightarrow$ uniformly continuous**
Assume not. Then some $\varepsilon_0 > 0$ and sequences $x_n, y_n \in [a, b]$ satisfy $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \ge \varepsilon_0$. Choose a subsequence $x_{n_k} \to x^* \in [a, b]$. Since $|x_{n_k} - y_{n_k}| \to 0$, also $y_{n_k} \to x^*$. Continuity gives $f(x_{n_k}), f(y_{n_k}) \to f(x^*)$, hence $|f(x_{n_k}) - f(y_{n_k})| \to 0$, a contradiction.

**Intermediate value theorem**
Assume $f(a) \le c \le f(b)$ and set $X := \{x \in [a, b] \mid f(x) \le c\}$, $\quad x^* := \sup X$. Choose $x_n \in X$ with $x_n \to x^*$; continuity gives $f(x^*) \le c$. If $f(x^*) < c$, continuity yields $\delta > 0$ such that $f(y) < c$ for $|y - x^*| < \delta$, producing $y > x^*$ in $X$, contradicting the definition of $x^*$. Hence $f(x^*) = c$.

**Cauchy criterion for series**
Let $s_n := \sum_{k=0}^{n} a_k$. Then $\sum_{k=0}^{\infty} a_k$ converges iff $(s_n)$ converges, iff $(s_n)$ is Cauchy. Since $s_n - s_m = \sum_{k=m+1}^{n} a_k$, this is equivalent to $\forall \varepsilon > 0\ \exists N\ \forall n \ge m \ge N : \left|\sum_{k=m+1}^{n} a_k\right| < \varepsilon$.

**Absolute convergence $\Rightarrow$ convergence**
Assume $\sum |a_n|$ converges. By the Cauchy criterion, for large $n \ge m$, $\sum_{k=m+1}^{n} |a_k| < \varepsilon$. Therefore $\left|\sum_{k=m+1}^{n} a_k\right| \le \sum_{k=m+1}^{n} |a_k| < \varepsilon$. Thus $\sum a_n$ is Cauchy and converges. Moreover, $\left|\sum_{k=0}^{n} a_k\right| \le \sum_{k=0}^{n} |a_k|$; passing to the limit gives $\left|\sum a_k\right| \le \sum |a_k|$.

**Leibniz criterion**
Let $s_n := \sum_{k=0}^{n} (-1)^k a_k$ with $a_n \downarrow 0$. Then $s_{2n+2} - s_{2n} = -a_{2n+1} + a_{2n+2} \le 0$, so $(s_{2n})$ decreases, while $s_{2n+3} - s_{2n+1} = a_{2n+2} - a_{2n+3} \ge 0$, so $(s_{2n+1})$ increases. Also $s_{2n+1} \le s_{2n}$ and $s_{2n} - s_{2n+1} = a_{2n+1} \to 0$. Thus both subsequences converge to the same limit, hence $(s_n)$ converges.

**Rearrangement of an absolutely convergent series**
Let $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ be bijective and $\sum |a_n| < \infty$. Given $\varepsilon > 0$, choose $N$ such that $\sum_{n>N} |a_n| < \varepsilon$. Choose $M$ so that $\{0, \ldots, N\} \subseteq \{\varphi(0), \ldots, \varphi(M)\}$. For $m \ge M$, the difference between $\sum_{k=0}^{m} a_{\varphi(k)}$ and $\sum_{n=0}^{\infty} a_n$ contains only terms with index $> N$, so its absolute value is $< \varepsilon$. Hence the rearranged series has the same sum; applying the same argument to $|a_n|$ gives absolute convergence.

# Solutions I

**Taylor polynomial**
Use $T_n f(x; x_0) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$. Here $x_0 = -1$ and $f(-1) = f'(-1) = f'''(-1) = 0$, $f''(-1) = 2$. Thus $T_2 f(x; -1) = \frac{2}{2!}(x + 1)^2 = (x + 1)^2$ and $T_3 f(x; -1) = (x + 1)^2$. Hence $f(-0.99) \approx T_2 f(-0.99; -1) = (0.01)^2 = 10^{-4}$.

**$x \sin(1/x)$ at 0**
Let $f(0) = 0$ and $f(x) = x \sin(1/x)$ for $x \ne 0$. The difference quotient is $\frac{f(h)-f(0)}{h} = \sin(1/h)$. Choose $h_n = 2/(\pi n) \to 0$; then $\sin(1/h_n) = \sin(\pi n/2)$ does not converge. Hence $f$ is not differentiable at 0. However, $|f(x)| \le |x| \to 0$, so $f$ is continuous at 0.

**$x^2 \sin(1/x)$ at 0**
Let $f(0) = 0$ and $f(x) = x^2 \sin(1/x)$ for $x \ne 0$. Then $\frac{f(h)-f(0)}{h} = h \sin(1/h)$ and $|h \sin(1/h)| \le |h| \to 0$. By the squeeze theorem, $\boxed{f'(0) = 0}$.

**Piecewise corner**
For $g(x) = e^{-x}$ if $x > 0$ and $g(x) = 1$ if $x \le 0$, $g$ is continuous at 0, but $g'_+(0) = \lim_{h\to 0^+} \frac{e^{-h}-1}{h} = -1$ and $g'_-(0) = \lim_{h\to 0^-} \frac{1-1}{h} = 0$. Thus $\boxed{g \text{ is not differentiable at } 0}$.

**Vertical tangent**
For $f(x) = \sqrt[3]{x}$, $\frac{f(x)-f(0)}{x} = x^{-2/3} \to +\infty$ as $x \to 0$. Thus no finite derivative exists at 0; geometrically the tangent is vertical.

**Rational–irrational function**
Let $f(x) = x$ for $x \in \mathbb{Q}$ and $f(x) = 0$ for $x \notin \mathbb{Q}$. At 0, $|f(x)| \le |x| \to 0$, so $f$ is continuous. For $x_0 \ne 0$, choose rational $r_n \to x_0$ and irrational $s_n \to x_0$. Then $f(r_n) \to x_0$ but $f(s_n) \to 0$. Hence $\boxed{f \text{ is continuous exactly at } 0}$.

**Power-series radius**
For $\sum a_n z^n$, $R = (\limsup_{n\to\infty} |a_n|^{1/n})^{-1}$, with $1/0 = \infty$. Thus $|z| < R$ gives absolute convergence and $|z| > R$ divergence. Examples: $a_n = (-1)^n/(2n + 1)$ gives $\limsup |a_n|^{1/n} = 1$, so $R = 1$; $a_n = e^{in}/(4n)!$ gives limit 0, so $R = \infty$; coefficients of order $1/k^2$ on even indices give $R = 1$.

**Rationalizing a root**
For $a_n \to 0$, $\frac{\sqrt{1+a_n}-1}{a_n} \cdot \frac{\sqrt{1+a_n}+1}{\sqrt{1+a_n}+1} = \frac{1}{\sqrt{1+a_n}+1}$. Since $\sqrt{1 + a_n} \to 1$, $\boxed{\dfrac{\sqrt{1 + a_n} - 1}{a_n} \to \dfrac{1}{2}}$.

**Spring oscillator**
At equilibrium, $-mg + ks_0 = 0 \Rightarrow s_0 = \frac{mg}{k}$. Let $x(t)$ be the displacement from equilibrium. Then Newton's law gives $m\ddot{x} + kx = 0$, $\quad \ddot{x} + \omega^2 x = 0$, $\quad \omega = \sqrt{\frac{k}{m}}$. The characteristic roots are $\lambda = \pm i\omega$, hence $x(t) = C_1 \sin(\omega t) + C_2 \cos(\omega t)$. For $x(0) = x_0$, $\quad \dot{x}(0) = v_0$, $\boxed{x(t) = \dfrac{v_0}{\omega} \sin(\omega t) + x_0 \cos(\omega t)}$. Moreover, $T = \frac{2\pi}{\omega}$, $\nu = \frac{\omega}{2\pi}$, $A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$. The energy $E(t) = \frac{m}{2}\dot{x}(t)^2 + \frac{k}{2}x(t)^2$ is conserved because $E'(t) = \dot{x}(t)\big(m\ddot{x}(t) + kx(t)\big) = 0$. Thus $\boxed{E(t) = E(0) = \dfrac{m}{2}v_0^2 + \dfrac{k}{2}x_0^2}$.

# Solutions II

**Subsequence criterion**
If $a_n \to L$ and $b_k = a_{n_k}$ is a subsequence, then $n_k \ge k$. For $\varepsilon > 0$, choose $N$ with $n \ge N \Rightarrow |a_n - L| < \varepsilon$. For $k \ge N$, $n_k \ge k \ge N$, hence $|b_k - L| = |a_{n_k} - L| < \varepsilon$. Thus $\boxed{b_k \to L}$.

**Dominant exponential**
For $\frac{(-2024)^n+(-2025)^n}{(-2024)^{n+1}-(-2025)^n}$, divide numerator and denominator by $(-2025)^n$: $\frac{(2024/2025)^n+1}{(2024/2025)^n(-2024)-1}$. Since $(2024/2025)^n \to 0$, the limit is $\boxed{-1}$.

**Root of exponential sum**
Since $17^n \le 5^n + 11^n + 17^n \le 3 \cdot 17^n$, taking $n$-th roots gives $17 \le \sqrt[n]{5^n + 11^n + 17^n} \le 17\sqrt[n]{3}$. As $\sqrt[n]{3} \to 1$, the squeeze theorem yields $\boxed{\sqrt[n]{5^n + 11^n + 17^n} \to 17}$.

**$n$-th root of a constant**
For $a > 1$, set $b_n = \sqrt[n]{a} - 1 \ge 0$. Bernoulli gives $1 + nb_n \le (1 + b_n)^n = a$, so $0 \le b_n \le (a - 1)/n \to 0$. Hence $\sqrt[n]{a} \to 1$. For $0 < a < 1$, apply the result to $1/a > 1$ and take reciprocals. Thus $\boxed{\lim_{n\to\infty} \sqrt[n]{a} = 1}$ for every $a > 0$.

**Polynomial roots**
For $p(x) = x^3 - 4x^2 + 9x - 10$, testing gives $p(2) = 0$. Polynomial division yields $p(x) = (x - 2)(x^2 - 2x + 5)$. Since $x^2 - 2x + 5 = (x - 1)^2 + 4 = (x - 1 - 2i)(x - 1 + 2i)$, the roots are $\boxed{2,\ 1 + 2i,\ 1 - 2i}$.

**Complex $n$-th roots**
Write $w = re^{i\varphi}$. The solutions of $z^n = w$ are $z_k = r^{1/n} e^{i(\varphi+2\pi k)/n}$, $k = 0, \ldots, n - 1$. For $64 + 64\sqrt{3}\,i = 128e^{i\pi/3}$ and $n = 7$, $\boxed{z_k = 2e^{i(\pi/21+2\pi k/7)}}$, $k = 0, \ldots, 6$.

**Trig identity via Euler**
Using $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$ and $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, expand $\sin x \cos y + \sin y \cos x$. After cancellation, $\frac{e^{i(x+y)}-e^{-i(x+y)}}{2i} = \sin(x + y)$. Hence $\boxed{\sin(x + y) = \sin x \cos y + \cos x \sin y}$.

**Affine supremum**
Let $M = \sup S$, $P = \sup\{ax + b : x \in S\}$ and $a > 0$. For $x \in S$, $x \le M \Rightarrow ax + b \le aM + b$, so $P \le aM + b$. Also, $ax + b \le P \Rightarrow x \le (P - b)/a$; thus $(P - b)/a$ is an upper bound of $S$, so $M \le (P - b)/a$ and $aM + b \le P$. Therefore $\boxed{\sup_{x\in S}(ax + b) = a \sup_{x\in S} x + b}$.

**Variable integral bound**
Use $\frac{d}{dx}\int_{g(x)}^{b} f(t)\,dt = -f(g(x))g'(x)$. Here $f(t) = \sin^2(t)\cos^2(t)$, $\quad g(x) = x^2$. Therefore $\boxed{G'(x) = -2x \sin^2(x^2)\cos^2(x^2)}$.

**Mixing tank**
Use rate of change = inflow − outflow. The salt inflow is $12 \cdot 0.125 = 1.5$ kg/min. Since the volume is constantly 400 l, the tank concentration is $\frac{c(t)}{400}$ kg/l. Thus the salt outflow is $12 \cdot \frac{c(t)}{400} = 0.03c(t)$ kg/min. Therefore $\boxed{c'(t) = 1.5 - 0.03c(t), \quad c(0) = 100}$.

**Population model**
Let $t$ be the number of years after 2010 and measure population in billions. The death rate is linear with $S(0) = 0.057$, $\quad S(30) = 0.080$. Hence $S(t) = 0.057 + \frac{0.080-0.057}{30}t = 0.057 + \frac{23}{30000}t$. The birth rate is $G(t) = 0.140$. Therefore $P'(t) = G(t) - S(t) = 0.083 - \frac{23}{30000}t$, $\quad P(0) = 6.9$. Integration gives $P(t) = 6.9 + 0.083t - \frac{23}{60000}t^2$. For 2050, $t = 40$, so $P(40) = 6.9 + 0.083 \cdot 40 - \frac{23}{60000} \cdot 40^2 \approx 9.61$. Thus $\boxed{P(2050) \approx 9.61 \text{ billion}}$.

**Complex polynomial roots**
Factor out the common factor: $p(z) = 2z^3 + (2 - 4i)z^2 - (6 + 2i)z = 2z(z^2 + (1 - 2i)z - (3 + i))$. Hence $z_0 = 0$. For the quadratic factor, $z_{1,2} = \frac{-(1-2i)\pm\sqrt{(1-2i)^2+4(3+i)}}{2}$. Since $(1 - 2i)^2 + 4(3 + i) = 9$, we get $z_{1,2} = \frac{-1+2i\pm 3}{2}$. Thus $z_1 = 1 + i$, $\quad z_2 = -2 + i$. Therefore $\boxed{p(z) = 2z(z - 1 - i)(z + 2 - i)}$ with zeros $\boxed{0,\ 1 + i,\ -2 + i}$.

**Optimal viewing distance**
For an object extending from height $a$ to $b$ with $0 < a < b$, the viewing angle at distance $d > 0$ is $\vartheta(d) = \arctan\left(\frac{b}{d}\right) - \arctan\left(\frac{a}{d}\right)$. Then $\vartheta'(d) = -\frac{b}{d^2+b^2} + \frac{a}{d^2+a^2}$. Setting $\vartheta'(d) = 0$ gives $a(d^2 + b^2) = b(d^2 + a^2) \iff d^2 = ab$. Since $\vartheta(d) \to 0$ for $d \to 0^+$ and $d \to \infty$, this critical point is the global maximum. Hence $\boxed{d = \sqrt{ab}}$. For $a = 46$, $b = 92$: $\boxed{d = \sqrt{46 \cdot 92} = 46\sqrt{2} \approx 65.05 \text{ m}}$.

**Differentiability of $|x|^3$**
Write $f(x) = |x|^3 = \begin{cases} x^3, & x \ge 0, \\ -x^3, & x < 0. \end{cases}$ For $x \ne 0$, both branches are differentiable. At $x = 0$, $f'_+(0) = \lim_{h\to 0^+} \frac{h^3}{h} = \lim_{h\to 0^+} h^2 = 0$, and $f'_-(0) = \lim_{h\to 0^-} \frac{-h^3}{h} = \lim_{h\to 0^-}(-h^2) = 0$. Since the one-sided derivatives agree, $\boxed{f \text{ is differentiable at } 0, \quad f'(0) = 0}$.